

Máster en Física Avanzada

Especialidad Física Teórica



Trabajo Fin de Máster

Fast Simulation of Neutrino Oscillations in Matter

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Curso académico 2025/26

Abstract

This Master’s Thesis presents **tpeanuts**, a fast, differentiable, GPU-batched framework for neutrino-oscillation phenomenology built on PyTorch and released as public, open-source software, with installation and usage documentation, for the neutrino-phenomenology community. Starting from a validated PyTorch translation of the PEANUTS toolkit’s numerical core, three-flavour oscillations, matter effects and solar-neutrino propagation, the project extends substantially: a modular architecture adds atmospheric-neutrino propagation through the Earth, several interchangeable solar and Earth matter/density models, beyond-the-Standard-Model scenarios, and a forward-folding detector layer coupled to a gradient-based statistical-inference engine that connects oscillation probabilities directly to published experimental data. Because its numerical core operates on batched PyTorch tensors, **tpeanuts** evaluates many oscillation configurations, energies or trajectories in parallel rather than one at a time, scales from CPU to GPU with no code changes, and, being differentiable end to end, connects directly to gradient-based inference and machine-learning applications, making it a common, reusable engine for solar, atmospheric and BSM neutrino studies alike.

The work reviews the theoretical basis of neutrino mixing, vacuum oscillations, the Mikheyev–Smirnov–Wolfenstein matter effect, non-standard interactions, sterile neutrinos and the statistical-inference formalism used throughout the analyses. It then describes the software strategy followed in **tpeanuts**: identification of the relevant PEANUTS modules, translation of numerical operations to tensor operations, preservation of units and conventions, construction of validation tests against the original implementation and against **nuSQuIDS**, and preparation of benchmarks for CPU and GPU execution.

The code is organised around a modular architecture in which solar, atmospheric and matter models, new-physics scenarios, detector responses and inference procedures can be selected and combined rather than hard-coded into a single implementation. Dedicated modules support beyond-the-Standard-Model physics, including non-standard interactions and sterile neutrinos. Built on top of this engine, a detector layer folds oscillated fluxes into predicted event counts for several real experiments, and an inference layer fits oscillation and detector parameters to their real, published data via automatic-differentiation-based likelihood minimisation. The most complete case, Daya Bay, is validated against the collaboration’s own published results: a near/far ratio fit recovers both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly, within 1–2% of Daya Bay’s published values. This establishes **tpeanuts** as a working foundation for real, gradient-based neutrino-oscillation data analysis, not only for fast forward simulation.

Keywords

neutrino oscillations; PEANUTS; PyTorch; differentiable programming; GPU computing; open-source software; MSW effect; solar neutrinos; atmospheric neutrinos; reactor antineutrinos; statistical inference.

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1 Introduction

Neutrino oscillations are one of the clearest experimental indications that the Standard Model of particle physics must be extended, as established by the atmospheric-neutrino measurements of Super-Kamiokande and the solar-neutrino measurements of the Sudbury Neutrino Observatory (Ahmad et al., 2002; Fukuda et al., 1998). A neutrino produced with a definite flavour, such as an electron, muon or tau neutrino, propagates as a coherent superposition of mass eigenstates. The interference among these components gives rise to flavour transitions whose probabilities depend on the neutrino energy, the travelled distance, the mixing parameters and, when propagation occurs in matter, the electron density of the medium (Giunti & Kim, 2007; Mikheyev & Smirnov, 1985; Wolfenstein, 1978).

Modern neutrino analyses require evaluating these probabilities many times. Oscillation probabilities appear inside flux predictions, detector simulations, likelihood scans and global fits. For this reason, the numerical implementation of the oscillation formalism is not a secondary detail: it directly affects the feasibility of high-statistics studies and of exploratory scans over large parameter spaces. Existing tools such as PEANUTS (Gonzalo & Lucente, 2024) provide compact routines for neutrino propagation, matter mixing and solar-neutrino probabilities. However, array-based implementations are not always optimal for current workflows involving GPUs, TPUs, automatic differentiation and batched execution.

This thesis studies and documents the development of **tpeanuts**, a PyTorch (Paszke et al., 2019)-based reproduction and extension of the PEANUTS neutrino-oscillation toolkit. The starting point is a validated, PyTorch-native translation of the core of PEANUTS, three-flavour vacuum and matter propagation and solar-neutrino probabilities, that preserves its physical conventions while expressing every operation as a tensor computation. This makes it possible to evaluate many oscillation configurations in parallel, to move computations between CPU and GPU with minimal changes, and to connect the resulting probabilities to gradient-based parameter estimation.

Built on top of this validated core, **tpeanuts** extends substantially PEANUTS: a modular architecture adds atmospheric-neutrino propagation through the Earth, several interchangeable solar and Earth matter/density models, beyond-the-Standard-Model scenarios (non-standard interactions and sterile neutrinos), and a full detector-level forward-folding and statistical-inference layer, validated against real, published data from several experiments. These extensions, together with the underlying oscillation engine, are released as a public, open-source Python package, with installation and usage documentation (section 5.6), so that other researchers can reuse and extend it directly rather than reimplementing it from the physics described here.

1.1 Motivation

The main motivation for **tpeanuts** is the need for fast, repeated neutrino-oscillation evaluation, both for large-scale simulation (atmospheric-flux studies, detector-response studies, BSM scans) and, ultimately, for fitting real data: a gradient-based fit to the published spectrum of a detector requires the oscillation probability, the detector response and the likelihood to be evaluated, and differentiated, at every optimiser step. PyTorch provides a natural environment for this because it combines vectorised tensor operations, GPU support and automatic differentiation, letting a

single implementation serve fast forward simulation and gradient-based inference alike.

The original PEANUTS implementation is tied to a specific solar model and a fixed treatment of matter effects, which limits its use where the sensitivity of a result to these choices must be tested. Building `tpeanuts` on a flexible tensor framework makes it possible to add such options, including alternative solar and Earth density models, perturbative treatments of the matter potential, and new-physics scenarios such as non-standard interactions (NSI) and sterile neutrinos, in a modular way, without compromising the validated core of the oscillation formalism. This same modularity is what makes it practical to add a detector-response layer (target composition, energy resolution, backgrounds, efficiencies) and an inference layer (likelihood construction, gradient-based fitting, uncertainty estimation) on top of the oscillation engine, and to reuse both across experiments as different as a reactor (Daya Bay), a liquid-scintillator solar detector (Borexino), a large-volume atmospheric detector (IceCube DeepCore) and a heavy-water Cherenkov detector (SNO).

Beyond the physics case for building it, three complementary features make `tpeanuts` useful to the wider neutrino-phenomenology community rather than only to this thesis. It is fast and scalable for large batched calculations: because the propagation engine operates on tensors rather than on scalars, thousands of oscillation configurations, spanning energies, baselines or points in a BSM parameter space, are evaluated in a single call instead of one at a time, with the advantage growing with batch size. It provides a common, flexible framework for solar, atmospheric and BSM neutrino studies that would otherwise require separate, purpose-built codes. And, being built on PyTorch, it is differentiable end to end, so oscillation probabilities can be connected directly to gradient-based inference and machine-learning applications rather than only to finite-difference or sampling-based methods. Combined with the fact that the code is public, on GitHub, with installation instructions and a per-module companion documentation set, this positions it as a reusable scientific tool rather than a single-purpose thesis codebase.

Delivering on this motivation is only credible if every result is validated, so this thesis validates `tpeanuts` along three complementary fronts. Internal validation checks the physics of the implemented neutrino-oscillation phenomena, verifying that independent numerical strategies for the same propagation problem agree with each other within the expected approximation. External validation compares `tpeanuts` against the original PEANUTS implementation and against `nuSQuIDS` (Argüelles et al., 2022), an independent, widely used neutrino-propagation toolkit, at the level of both intermediate quantities and final probabilities. Finally, established experimental and observational results from the literature, from the solar day–night asymmetry to real detector data, are reproduced directly. Combining these three fronts is what makes it possible to obtain a tool that is simultaneously fast, reliable and versatile.

1.2 Objectives

The general objective of this Master’s Thesis is to develop and document `tpeanuts` as a modular, extensible and publicly available PyTorch framework for neutrino-oscillation phenomenology, built on a validated translation of the core of PEANUTS and extended into a detector-level statistical-inference framework validated against real experimental data. The specific objectives are:

- to review the three-flavour oscillation formalism used by PEANUTS, including vacuum

propagation and matter effects, and to translate its numerical core to PyTorch tensors while preserving physical conventions, units and sign choices;

- to design a modular architecture in which vacuum/solar/Earth/atmosphere models, matter and perturbative treatments, and new-physics scenarios (NSI, sterile neutrinos) can be selected and reused consistently across the application;
- to define a validation strategy comparing **tpeanuts** against PEANUTS and **nuSQuIDS** at the level of intermediate quantities and final probabilities, complemented by the reproduction of well-known theoretical and experimental results and by a CPU/GPU benchmarking methodology;
- to implement a detector-agnostic forward-folding layer (cross sections, target composition, energy response, efficiency, backgrounds) and to specialise it for real experiments;
- to implement a gradient-based statistical-inference layer (Poisson and Gaussian likelihoods, LBFGS fitting, Laplace/Fisher uncertainty estimation, likelihood scans) built on PyTorch autograd;
- to validate the full detector-inference chain against real, published data, with Daya Bay as the primary case study, and to document its application to Borexino, IceCube DeepCore and SNO;
- to release **tpeanuts** publicly, as a documented, installable, open-source package intended for reuse by the neutrino-phenomenology community, rather than as a closed thesis-specific codebase.

1.3 Document structure

Section 2 introduces the theoretical framework of neutrino masses, mixing, oscillations, matter effects and the statistical-inference formalism. Section 3 describes the methodology used to translate and validate the numerical core, extend it into a modular propagation framework and into beyond-the-Standard-Model scenarios, validate the resulting inference chain against real data, and release the code publicly. Section 4 presents the technical development of **tpeanuts**, including its detector and inference layers. Section 5 first establishes the numerical reliability and computational usefulness of the implementation through validation and benchmarking against PEANUTS and **nuSQuIDS**, then demonstrates capabilities that go beyond a translation of PEANUTS, atmospheric-neutrino propagation through the Earth and flexible matter/density models, followed by the NSI and sterile-neutrino extensions, then the phenomenological application of the full framework to real experimental data, headlined by the Daya Bay near/far detector-level case study, and closes with the public repository underlying every result. Finally, Section 6 summarises the conclusions and outlines future work.

2 Theoretical Framework

This section summarises the physical ingredients required by **tpeanuts**. The objective is not to provide a complete review of neutrino physics, but to establish the conventions that must be preserved when updating PEANUTS to a tensor-based implementation.

2.1 Neutrino propagation fundamentals

This subsection collects the formalism common to every propagation regime considered in this thesis: the definition of the mixing matrix, vacuum oscillations, and the modification of the propagation Hamiltonian by ordinary matter. The specific neutrino sources and new-physics extensions built on top of this common core are treated separately in the following subsections.

2.1.1 Neutrino mixing

One of the fundamental discoveries in particle physics is that neutrinos are produced and detected in flavour eigenstates, whereas they propagate through space as mass eigenstates. Since these two bases do not coincide, a neutrino created with a definite flavour evolves into a quantum superposition of different flavours during propagation, giving rise to the phenomenon of neutrino oscillations (Giunti & Kim, 2007).

Within the Standard Model extended to include neutrino masses, the flavour eigenstates $|\nu_\alpha\rangle$, $\alpha = e, \mu, \tau$, are related to the mass eigenstates $|\nu_i\rangle$, $i = 1, 2, 3$, through a unitary transformation known as the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix (Maki et al., 1962; Pontecorvo, 1957, 1958),

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle, \quad (1)$$

or, equivalently,

$$|\nu_i\rangle = \sum_{\alpha=e,\mu,\tau} U_{\alpha i} |\nu_\alpha\rangle. \quad (2)$$

The matrix U is unitary, $U^\dagger U = U U^\dagger = \mathbb{I}$, which guarantees the conservation of probability during neutrino propagation.

Physical interpretation. The matrix elements $U_{\alpha i}$ represent the probability amplitudes that a neutrino with flavour α is found in the mass eigenstate i . Consequently, $|U_{\alpha i}|^2$ gives the probability that a neutrino produced as flavour α propagates in the mass eigenstate i . This mismatch between flavour and mass eigenstates is directly analogous to the quark sector, where flavour mixing is described by the Cabibbo–Kobayashi–Maskawa (CKM) matrix (Workman et al., 2022). In the lepton sector, however, the mixing angles are significantly larger, making flavour oscillations observable over laboratory, atmospheric, solar and astrophysical baselines.

Standard parametrisation. For three active Dirac neutrinos, the PMNS matrix is written as the product of three elementary rotations, $U_{\text{PMNS}} = R_{23}(\theta_{23}) U_{13}(\theta_{13}, \delta_{\text{CP}}) R_{12}(\theta_{12})$, yielding the standard Particle Data Group (PDG) parametrisation (Workman et al., 2022)

$$U_{\text{PMNS}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{\text{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{23}c_{13} \end{pmatrix}, \quad (3)$$

with $c_{ij} = \cos \theta_{ij}$, $s_{ij} = \sin \theta_{ij}$. Each rotation has a clear physical meaning: θ_{12} governs mixing between the first and second mass eigenstates and dominates solar-neutrino oscillations

together with the smaller mass splitting Δm_{21}^2 ; θ_{23} controls mixing between the second and third eigenstates and is primarily responsible for atmospheric-neutrino oscillations; θ_{13} couples the first and third eigenstates and, although considerably smaller than the other mixing angles, plays a crucial role in matter effects and enables sensitivity to CP violation. The complex phase δ_{CP} introduces a difference between neutrino and antineutrino oscillation probabilities, providing a possible source of CP violation in the lepton sector.

Oscillation parameters. Oscillation experiments are sensitive not to the absolute neutrino masses but to the differences between their squared masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Only two independent mass-squared differences exist, Δm_{21}^2 and Δm_{31}^2 (or equivalently Δm_{32}^2), together with the three mixing angles and the CP-violating phase,

$$\{\theta_{12}, \theta_{13}, \theta_{23}, \Delta m_{21}^2, \Delta m_{31}^2, \delta_{\text{CP}}\}. \quad (4)$$

These six quantities completely determine neutrino oscillation probabilities within the standard three-neutrino oscillation framework, with current best-fit values obtained through global analyses combining solar, atmospheric, reactor and accelerator data. The NuFIT collaboration's NuFIT 5.2 release (Esteban et al., 2020) fixes the oscillation-parameter presets used throughout this work's forward-simulation and validation notebooks; the more recent NuFIT 6.1 update (Esteban et al., 2025), based on the same underlying analysis (Esteban et al., 2024), is used separately, only as an external benchmark in the real-data detector-level fits of section 3.4, not as a simulation input.

2.1.2 Vacuum oscillations

Once produced in a flavour eigenstate, neutrinos propagate in vacuum as coherent superpositions of mass eigenstates. Since each mass eigenstate accumulates a different quantum phase, their interference is the physical origin of neutrino oscillations (Giunti & Kim, 2007). The evolution is governed by the Schrödinger equation,

$$i \frac{d}{dt} |\nu(t)\rangle = H |\nu(t)\rangle, \quad (5)$$

where H is the effective Hamiltonian describing neutrino propagation.

Vacuum Hamiltonian. In vacuum, H is diagonal in the mass basis, $H_{\text{mass}} = \text{diag}(E_1, E_2, E_3)$, with $E_i = \sqrt{p^2 + m_i^2}$. Since neutrinos produced in laboratory, solar and atmospheric experiments are highly relativistic, $m_i^2 \ll p^2$, the energy can be expanded as $E_i \simeq p + m_i^2/(2E)$, with $E \simeq p$ the neutrino energy. The common momentum term contributes only an overall global phase, e^{-ipL} , which has no observable consequences and can be dropped, leaving the effective vacuum Hamiltonian

$$H_{\text{vac}} = \frac{1}{2E} \text{diag}(m_1^2, m_2^2, m_3^2). \quad (6)$$

Flavour evolution requires transforming H_{vac} into the flavour basis through the PMNS matrix,

$$H_f = U H_{\text{vac}} U^\dagger = \frac{1}{2E} U \text{diag}(m_i^2) U^\dagger. \quad (7)$$

Evolution operator. For a constant Hamiltonian, the Schrödinger equation is solved by $|\nu(L)\rangle = e^{-iHL}|\nu(0)\rangle$. For a neutrino produced in flavour α , $|\nu(0)\rangle = |\nu_\alpha\rangle$, the transition amplitude to flavour β after a distance L is

$$A_{\alpha\rightarrow\beta}(L, E) = \sum_{i=1}^3 U_{\beta i} \exp\left(-i\frac{m_i^2 L}{2E}\right) U_{\alpha i}^*, \quad (8)$$

with oscillation probability $P_{\alpha\rightarrow\beta} = |A_{\alpha\rightarrow\beta}|^2$. Unitarity of the PMNS matrix guarantees $\sum_\beta P_{\alpha\rightarrow\beta} = 1$, i.e. conservation of probability.

Oscillation phases. Oscillations arise from the phase differences accumulated by the mass eigenstates, controlled by the mass-squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$, which define the phase entering the transition amplitude of equation (8), $\Delta_{ij} = \Delta m_{ij}^2 L / 2E$. Squaring a coherent two-state superposition introduces a half-angle: the two-flavour reduction of equation (8) gives $P = 1 - \sin^2(2\theta) \sin^2(\Delta_{ij}/2)$, so it is $\Delta_{ij}/2$, not Δ_{ij} itself, that governs the observable probability. In the practical units customary in oscillation phenomenology (Workman et al., 2022), $\Delta_{ij}/2 = 1.267 \Delta m_{ij}^2 [\text{eV}^2] L [\text{km}] / E [\text{GeV}]$; oscillations are thus governed by the ratio L/E rather than by the baseline or the energy separately.

2.1.3 Matter effects

The vacuum formalism above assumes that neutrinos propagate through empty space. In many settings relevant to **tpeanuts**, such as the solar interior, the Earth’s mantle and core, or a supernova envelope, neutrinos instead traverse dense media, where coherent forward scattering off the particles of the medium modifies the propagation Hamiltonian. This is the matter effect, also known as the Mikheyev–Smirnov–Wolfenstein (MSW) effect (Giunti & Kim, 2007; Mikheyev & Smirnov, 1985; Wolfenstein, 1978). Because coherent forward scattering transfers essentially no momentum to the medium, it does not remove neutrinos from the beam; instead, it modifies the phase accumulated during propagation, which is described as an additional effective potential in the Schrödinger equation.

Matter potential. Neutrinos interact with matter through both neutral-current (NC) and charged-current (CC) weak processes. NC interactions, $\nu_\alpha + f \rightarrow \nu_\alpha + f$ with f an electron, proton or neutron, affect all active flavours equally; since oscillation probabilities depend only on relative phases between flavours, this common contribution is unobservable and is omitted from the oscillation Hamiltonian. Electron neutrinos additionally undergo the CC process $\nu_e + e^- \rightarrow \nu_e + e^-$, mediated by $W^\pm$ exchange, which introduces a flavour-dependent potential

$$V_{\text{CC}} = \sqrt{2} G_F n_e, \quad (9)$$

where G_F is the Fermi coupling constant and n_e the local electron number density. This additional, electron-flavour-specific interaction breaks the flavour symmetry of vacuum propagation and is the origin of the MSW effect.

Effective Hamiltonian. The full Hamiltonian governing flavour evolution in matter is obtained by adding the matter potential to the vacuum Hamiltonian,

$$H = H_{\text{vac}} + V_{\text{mat}}, \quad H_{\text{vac}} = \frac{1}{2E} U \text{diag}(m_1^2, m_2^2, m_3^2) U^\dagger, \quad V_{\text{mat}} = \sqrt{2} G_F n_e \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (10)$$

where V_{mat} is diagonal in the flavour basis, since only the electron flavour experiences the CC interaction. For antineutrinos, both the matter potential changes sign, $V_{\text{mat}} \rightarrow -V_{\text{mat}}$, and the mixing matrix is complex-conjugated, $U \rightarrow U^*$ (equivalently $\delta_{\text{CP}} \rightarrow -\delta_{\text{CP}}$), a convention detailed in section 4.2.2. In matter, the propagation basis becomes density-dependent, $H = U_m \text{diag}(\lambda_1, \lambda_2, \lambda_3) U_m^\dagger$, with U_m the effective mixing matrix and λ_i the effective eigenvalues; as the electron density varies along the trajectory, both evolve continuously, responsible for the characteristic energy dependence of solar-neutrino oscillations.

2.2 Neutrino sources

The oscillation formalism introduced above is universal: the same Schrödinger equation governs flavour evolution regardless of how the neutrino is produced. What varies between sources is the physical environment of production and, consequently, the density profile traversed before detection, which determines the relative importance of vacuum oscillations, matter effects and decoherence. Each source is therefore sensitive to a different region of the oscillation parameter space.

The current version of **tpeanuts** is designed primarily for the simulation of solar and atmospheric neutrinos, which together probe a broad range of energies, baselines and matter densities. Owing to its modular architecture, however, the propagation engine itself is independent of the neutrino source, so that additional sources can be incorporated with minimal modifications, as discussed for reactor and accelerator neutrinos below.

2.2.1 Solar neutrinos

Solar neutrinos are produced by thermonuclear fusion in the solar core, where temperatures reach $\sim 1.5 \times 10^7$ K and densities exceed 150 g cm^{-3} (Vinyoles et al., 2017). The dominant production mechanisms are the proton–proton (pp) chain and the CNO cycle, generating neutrinos from a few hundred keV up to ~ 20 MeV (Vinyoles et al., 2017).

Unlike reactor or accelerator neutrinos, solar neutrinos are produced deep inside a medium whose density decreases continuously from the core to the photosphere, so that flavour evolution is strongly affected by the MSW effect. For the Large Mixing Angle (LMA) solution favoured by current oscillation data (Esteban et al., 2020), propagation through the Sun is highly adiabatic over most of the relevant energy range: the neutrino remains in the same instantaneous matter eigenstate as the local density changes (Haxton, 1986). Within this approximation, the flavour state produced at radius r is projected onto the local matter eigenstates, which evolve adiabatically into vacuum mass eigenstates as the neutrino exits the Sun. After propagating over the Earth–Sun distance (1 AU), the mass eigenstates lose mutual coherence due to wave-packet separation and the finite energy resolution of detectors (Giunti & Kim, 2007), so the flavour transition probabilities reduce to an incoherent sum,

$$P_{\nu_e \rightarrow \nu_\alpha}^{(f)}(E) = \sum_{i=1}^3 |U_{\alpha i}|^2 T_i^{(f)}(E), \quad (11)$$

where $T_i^{(f)}(E)$ is the probability that a neutrino produced through solar reaction f emerges from the Sun as vacuum mass eigenstate ν_i . Since production is spatially distributed throughout the solar interior, $T_i^{(f)}$ is obtained by averaging over the corresponding radial production profile,

$$T_i^{(f)}(E) = \int_0^{R_\odot} w_f(r) P_{\nu_e \rightarrow \nu_i}(E, r) dr, \quad (12)$$

with $w_f(r)$ the normalised production distribution predicted by the Standard Solar Model (SSM). The solar module in **tpeanuts** uses radial electron-density profiles together with production distributions for the principal channels (pp , pep , hep , ${}^7\text{Be}$, ${}^8\text{B}$, ${}^{13}\text{N}$, ${}^{15}\text{O}$, ${}^{17}\text{F}$), using standard solar models such as B16(GS98) and B16(AGSS09met) (Vinyoles et al., 2017). Optional non-adiabatic corrections based on the Landau–Zener formalism (Parke, 1986) are also available for the regions where the adiabatic approximation becomes less accurate. Because solar neutrinos experience both matter-enhanced conversion inside the Sun and, at night, regeneration inside the Earth, they are one of the most powerful probes of neutrino propagation in matter and provide stringent constraints on the solar mixing parameters and on possible non-standard interactions.

2.2.2 Atmospheric neutrinos

Atmospheric neutrinos are produced when high-energy cosmic rays interact with nuclei in the atmosphere, generating hadronic cascades dominated by pions and kaons; the decay of these mesons and of the resulting secondary muons produces both muon and electron neutrinos, spanning energies from a few hundred MeV to beyond the TeV scale (Honda et al., 2007). Unlike solar neutrinos, they are produced over a wide range of altitudes and arrive at terrestrial detectors from all directions, with baselines extending from ~ 20 km for downward-going events to nearly the Earth’s diameter ($\sim 12\,742$ km) for upward-going neutrinos that cross the planet.

This broad coverage in energy and baseline makes atmospheric neutrinos sensitive to several oscillation regimes at once: downward-going events are dominated by atmospheric or vacuum oscillations, while upward-going neutrinos experience significant matter effects while traversing the Earth’s mantle and core. The interplay between the two regimes produces the characteristic oscillation patterns in the $(E, \cos \theta_z)$ plane known as oscillograms. Within **tpeanuts**, atmospheric-neutrino calculations combine the three-flavour propagation formalism with realistic Earth density profiles, such as the Preliminary Reference Earth Model (PREM) (Dziewonski & Anderson, 1981), and differential atmospheric flux tables from the Honda parametrisation (Honda et al., 2007) or from cascade solvers such as MCEq (Fedynitch et al., 2015).

2.2.3 Reactor antineutrinos

Nuclear reactors are intense, well-characterised sources of electron antineutrinos, emitted in the beta decays of the neutron-rich fission fragments of ${}^{235}\text{U}$, ${}^{238}\text{U}$, ${}^{239}\text{Pu}$ and ${}^{241}\text{Pu}$, with a spectrum extending up to ~ 8 MeV and a rate set by the reactor’s thermal power. Because flavour evolution is entirely determined by the vacuum Hamiltonian of section 2.1, and reactor baselines ($\lesssim 2$ km for a near/far experiment such as Daya Bay) traverse negligible Earth matter,

reactor antineutrinos follow the exact coherent three-flavour vacuum survival probability, with θ_{13} governing $\bar{\nu}_e \rightarrow \bar{\nu}_e$ disappearance together with the atmospheric-scale splitting. At this baseline $\sin^2 \Delta_{31}$ and $\sin^2 \Delta_{32}$ are numerically close enough, differing only through the small Δ_{21} phase, that what the disappearance channel directly resolves is the effective, ordering-insensitive combination $\Delta m_{ee}^2 = \cos^2 \theta_{12} \Delta m_{31}^2 + \sin^2 \theta_{12} \Delta m_{32}^2$, rather than Δm_{31}^2 itself, which is only recovered after fixing θ_{12} and Δm_{21}^2 externally (section 5.5.1). **tpeanuts** implements a full reactor-antineutrino observable chain on top of this probability, from the per-isotope Huber–Mueller emission spectrum (Huber, 2011; Mueller et al., 2011) through the inverse-beta-decay cross section (Vogel & Beacom, 1999) to a detector-response model folding the true antineutrino spectrum into an observable prompt-energy spectrum, described in section 4.1.1 and validated against real Daya Bay data in section 5.5.1.

2.2.4 Other neutrino sources

Accelerator-based experiments produce controlled muon-neutrino or antineutrino beams over baselines from a few hundred to more than a thousand kilometres, enabling precision measurements of the atmospheric oscillation parameters, the CP-violating phase and the neutrino mass ordering. As with reactor antineutrinos, incorporating such a source requires only specifying the appropriate initial flavour composition, energy spectrum, baseline and matter-density profile within the same propagation engine, without any change to the oscillation formalism itself.

2.3 New physics

Beyond-the-Standard-Model scenarios can be incorporated into the same formalism through modifications of the effective Hamiltonian and, in the case of sterile neutrinos, of the PMNS matrix itself. Two such extensions are considered in **tpeanuts**: non-standard interactions and sterile neutrinos.

2.3.1 Non-standard interactions (NSI)

Although the three-flavour framework introduced above explains the vast majority of neutrino oscillation observations, it is widely regarded as an effective low-energy description rather than a complete theory: the Standard Model does not account for the origin of neutrino masses, nor does it explain the nature of dark matter, the matter–antimatter asymmetry or the hierarchy of fermion masses. Many theories beyond the Standard Model (BSM) therefore predict additional neutrino interactions that would manifest as small deviations from the standard oscillation paradigm.

Among the most studied such scenarios are non-standard interactions (NSI), in which neutrinos experience additional weak-scale interactions with matter mediated by new heavy particles (Grossman, 1995). At energies well below the mediator mass scale, these effects can be described with an effective field theory in which the heavy degrees of freedom are integrated out and encoded in dimensionless effective couplings, providing a model-independent framework for a broad class of BSM scenarios without specifying the underlying microscopic theory (Farzan & Tortola, 2018).

NSI can in general affect neutrino production, detection and propagation; this work considers only *propagation* NSI, in which new neutral-current-like interactions modify the coherent forward scattering of neutrinos through matter. Since coherent forward scattering only changes

the propagation phase, these interactions appear as an additional contribution to the matter Hamiltonian,

$$H_{\text{mat}} = \sqrt{2}G_F n_e \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \varepsilon_{ee} & \varepsilon_{e\mu} & \varepsilon_{e\tau} \\ \varepsilon_{e\mu}^* & \varepsilon_{\mu\mu} & \varepsilon_{\mu\tau} \\ \varepsilon_{e\tau}^* & \varepsilon_{\mu\tau}^* & \varepsilon_{\tau\tau} \end{pmatrix} \right], \quad (13)$$

where the dimensionless couplings $\varepsilon_{\alpha\beta}$, normalised to the electron density n_e for a given medium composition, quantify the interaction strength relative to the Standard Model weak interaction. Only the traceless part of the diagonal is observable, since a common shift to all three entries is an unobservable global phase; **tpeanuts** adopts the standard convention $\varepsilon_{\mu\mu} = 0$. The off-diagonal terms introduce flavour-changing coherent interactions with no Standard Model analogue; for antineutrinos, $\varepsilon_{\alpha\beta}$ is complex-conjugated together with the sign flip of section 4.2.2. More generally, NSI couple to the individual matter fermions e , u , d rather than to a single medium, so the coupling above is really $\varepsilon_{\alpha\beta}(x) = \varepsilon_{\alpha\beta}^{(e+p)} + (n_n(x)/n_e(x))\varepsilon_{\alpha\beta}^n$, using electrical neutrality ($n_p = n_e$) to collapse e, u, d into two effective pieces that **tpeanuts** implements separately, ε^n defaulting to zero to recover equation (13); this matters because the Sun and the Earth have different neutron-to-electron ratios, one ingredient of the LMA-Dark degeneracy below. The complete Hamiltonian is $H = H_{\text{vac}} + H_{\text{mat}}$, with $H_{\text{vac}} = \frac{1}{2E}UM^2U^\dagger$ unchanged: NSI leave vacuum oscillations unaffected but modify matter propagation, with an impact depending on density, composition and energy, making long-baseline, atmospheric and solar experiments particularly sensitive.

Phenomenologically, NSI modify the effective mixing angles and mass splittings in matter, shift the MSW resonance, alter the day–night regeneration of solar neutrinos, and can introduce degeneracies with the standard oscillation parameters. A well-known example is the LMA-Dark solution (Esteban et al., 2018; Miranda et al., 2006), where a large negative ε_{ee} compensates a transformation of the solar mixing angle, yielding oscillation probabilities nearly indistinguishable from the standard Large Mixing Angle (LMA) solution; such degeneracies illustrate the value of combining different classes of neutrino experiments to disentangle standard oscillation physics from possible BSM effects. Current data constrain most individual NSI couplings to be subdominant with respect to the Standard Model weak interaction, but not universally: near degeneracies such as LMA-Dark, large couplings survive precisely because they are compensated by a parameter transformation rather than excluded outright, and couplings involving the τ flavour remain only weakly bounded on their own; global analyses combining oscillation, scattering and collider measurements give increasingly stringent, but data-set-dependent, limits (Biggio et al., 2009; Esteban et al., 2018).

From the computational point of view, NSI are a natural extension of the propagation formalism implemented in **tpeanuts**: since they modify only the matter contribution to the Hamiltonian, the numerical propagation algorithms developed for the standard three-flavour model remain unchanged. The implementation introduces a fully general complex Hermitian NSI matrix, allowing both flavour-conserving and flavour-changing couplings, which enables direct comparison between Standard Model predictions and a broad range of BSM scenarios while preserving the efficiency and flexibility of the underlying PyTorch implementation.

2.3.2 Sterile neutrinos

The Standard Model contains three active flavours, ν_e , ν_μ and ν_τ , all of which participate in the weak interaction through charged- and neutral-current processes. Several BSM models, together with a number of experimental anomalies reported over the past decades, have nonetheless motivated the existence of additional neutrino states that are singlets under the Standard Model gauge group. Because they do not couple directly to the weak interaction, these hypothetical states are known as *sterile* neutrinos (Böser et al., 2020). Although they cannot be produced or detected through Standard Model weak processes, sterile neutrinos may mix with the active flavours through the neutrino mass matrix and thus participate in flavour oscillations as additional propagation eigenstates, modifying the observed oscillation pattern without introducing any new interaction at the detector.

The simplest and most widely studied extension is the $3 + 1$ model (Giunti & Lasserre, 2019; Kopp et al., 2013), in which a single sterile state is added to the three active flavours. The flavour basis becomes $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$ and the propagation basis four mass eigenstates $(\nu_1, \nu_2, \nu_3, \nu_4)$, enlarging the 3×3 PMNS matrix into a 4×4 unitary mixing matrix, built here as

$$U_4 = R_{23} \Delta R_{24}(\theta_{24}, \delta_{24}) R_{34}(\theta_{34}, \delta_{34}) R_{13} \Delta^\dagger R_{12} R_{14}(\theta_{14}, \delta_{14}), \quad (14)$$

extending the three-flavour factorisation of section 2.1, with R_{23} , Δ , R_{13} , R_{12} embedded unchanged and three new active-sterile rotations R_{14} , R_{24} , R_{34} , each carrying its own CP phase. This ordering is **tpeanuts**'s own implementation convention rather than a universal standard, and must be respected exactly: R_{13} and R_{34} share the τ index, and Δ fails to commute with R_{34} once $\delta_{\text{CP}} \neq 0$, so it is not equivalent to a block-embedded U_{PMNS} followed by the sterile rotations. Beyond the three-flavour parameters, the $3 + 1$ framework introduces three new mixing angles $\theta_{14}, \theta_{24}, \theta_{34}$; **tpeanuts** parametrises the sterile sector with three raw phases $\delta_{14}, \delta_{24}, \delta_{34}$, of which only two are physically independent, consistent with the general $N \times N$ count of $(N - 1)(N - 2)/2$ physical phases (three for $N = 4$, including δ_{CP}). The extension also adds a fourth mass eigenstate m_4 and $\Delta m_{41}^2 = m_4^2 - m_1^2$, typically $\gg |\Delta m_{31}^2|$ (of order 1 eV^2) in short-baseline-anomaly phenomenology (Kopp et al., 2013), giving oscillation lengths much shorter than the atmospheric or solar ones.

Unlike NSI, which modify only the matter contribution, sterile neutrinos change the formalism more fundamentally: the vacuum Hamiltonian itself must be enlarged. The formalism of section 2.1 is expressed in natural units, $\hbar = c = 1$; **tpeanuts**'s internal normalisation rescales it by a purely numerical reference length L_{scale} , together with the explicit $\hbar c$ conversion needed once L_{scale} is expressed in metres rather than natural (inverse-energy) units, without changing the underlying physics. Consistently with the matter potentials below, the enlarged vacuum term is

$$H_{\text{vac}} = \frac{L_{\text{scale}}}{2E\hbar c} U_4 \text{diag}(m_1^2, m_2^2, m_3^2, m_4^2) U_4^\dagger, \quad (15)$$

so the new eigenstate contributes its own phase, $\exp(-im_4^2 L/2E)$ in natural units, to the evolution operator. Matter effects are modified too: the sterile state feels neither the charged- nor the neutral-current (NC) potential common to the three active flavours, $V_{\text{NC}} = \mp \frac{1}{\sqrt{2}} G_F n_n L_{\text{scale}}$

(from the neutron density n_n) — an unobservable global phase in the pure three-flavour case, but a genuine, observable relative potential once this NC-blind sterile state is added: after subtracting the common phase,

$$H_{\text{mat}} = \text{diag}(V_{\text{CC}}, 0, 0, -V_{\text{NC}}), \quad (16)$$

with $V_{\text{CC}} = \sqrt{2}G_F n_e L_{\text{scale}}$. This NC term is an optional addition, active whenever a neutron-density profile is supplied; omitting it recovers the CC-only sterile matter term used as the default here, an approximation once θ_{24} or θ_{34} are non-zero, and one not to be assumed negligible in general: the NC potential is standard practice for matter-affected active-sterile propagation in the literature, underlying the Earth-matter resonance exploited by atmospheric and long-baseline ν_μ - ν_s searches (IceCube Collaboration, Abbasi, et al., 2020). Either way, the modified matter Hamiltonian alters the effective eigenvalues and can generate additional resonant conversion in dense media.

Phenomenologically, sterile neutrinos may produce anomalous flavour disappearance, unexpected appearance channels, distortions of oscillation spectra and modified matter-induced effects, and have been extensively investigated in connection with the LSND and MiniBooNE anomalies, the reactor antineutrino anomaly and the Gallium anomaly, although no conclusive evidence has yet established their existence (Böser et al., 2020). Cosmological observations independently constrain light sterile neutrinos through their impact on Big Bang nucleosynthesis, the effective number of relativistic degrees of freedom, and large-scale structure formation.

Within **tpeanuts**, sterile neutrinos are implemented as a natural extension of the standard three-flavour propagation framework: the oscillation engine has been generalised to support arbitrary Hamiltonian dimensions, so that the 4×4 mixing matrix, the enlarged mass spectrum and the corresponding matter Hamiltonian are treated with the same numerical propagation algorithms developed for the standard model. The current implementation fully supports the $3 + 1$ scenario, while its modular design allows straightforward extension to models with multiple sterile states or more general BSM oscillation frameworks.

2.4 Statistical inference formalism

Comparing an oscillation prediction against real, binned experimental data requires a statistical framework connecting predicted and observed counts, a rule for minimising the resulting statistic with respect to the oscillation and detector parameters, and a way of estimating the uncertainty on the fitted values. **tpeanuts** implements this framework directly on top of PyTorch’s automatic differentiation, so that the same gradient used to build the oscillation Hamiltonian also drives the fit.

For a set of bins with predicted counts $N_k(\theta)$ and observed counts n_k^{obs} , the Poisson (Baker–Cousins) likelihood-ratio statistic (Baker & Cousins, 1984)

$$-2 \ln L_{\text{Poisson}} = 2 \sum_k \left[N_k(\theta) - n_k^{\text{obs}} + n_k^{\text{obs}} \ln \frac{n_k^{\text{obs}}}{N_k(\theta)} \right] \quad (17)$$

is used for absolute event counts, and vanishes exactly when $N_k = n_k^{\text{obs}}$ bin by bin. For derived observables that are not themselves Poisson-distributed, such as a ratio of counts, a Gaussian

statistic with propagated uncertainty is used instead, $-2 \ln L_{\chi^2} = \sum_k [(R_k(\theta) - R_k^{\text{obs}})/\sigma_k]^2$. Externally measured nuisance parameters (background normalisations, energy-scale pulls) are constrained by an additional Gaussian penalty term on θ itself, added to either statistic.

The best-fit point $\hat{\theta} = \arg \min_{\theta} [-2 \ln L(\theta)]$ is found with the quasi-Newton LBFGS optimiser, using the exact gradient of the total statistic obtained by backpropagating through the full physics chain, oscillation probability, detector response and likelihood, rather than finite differences. The uncertainty on $\hat{\theta}$ is estimated through the Laplace approximation, in which the Fisher information at the optimum, $\mathcal{I} = \frac{1}{2} \nabla_{\theta}^2 (-2 \ln L)$, obtained as the autograd Hessian of the statistic, gives the parameter covariance $\Sigma = \mathcal{I}^{-1}$, with $\sigma_j = \sqrt{\Sigma_{jj}}$. Where the local Laplace approximation is inadequate, in particular along directions the data barely constrain, a direct two-dimensional scan of $-2 \ln L$ over a parameter grid, optionally profiling the remaining free parameters at each grid point, provides confidence contours at Wilks'-theorem asymptotic thresholds, valid under standard regularity conditions but liable to fail near physical boundaries, degeneracies or low statistics; since **tpeanuts** performs no Monte Carlo (e.g. Feldman–Cousins) coverage calibration, these contours should be read as asymptotic, not empirically calibrated.

2.5 Related computational tools

tpeanuts interfaces with several public tools, either for cross-validation or as external inputs. **nuSQuIDS** (Arguelles et al., 2022), a general framework for solving neutrino quantum kinetic equations, is used as an independent, actively maintained numerical solver against which vacuum, solar and atmospheric probabilities are cross-validated; **PEANUTS** remains lighter and more specialised, precisely the property **tpeanuts** aims to preserve. **MCEq** (Fedynitch et al., 2015), which solves the coupled cascade equations of cosmic-ray-induced air showers, and the **Honda/HKKM** tables (Honda et al., 2007) provide the two atmospheric-neutrino flux inputs; the companion **CRFlux** library supplies the primary cosmic-ray flux parametrisations (e.g. Polygonato) shared by **MCEq** and the **CORSIKA**-based sampling pipeline, so that both remain anchored to the same top-of-atmosphere spectrum. **pymis** (Emmert et al., 2021) provides empirical atmospheric density/composition profiles as a function of altitude, location and time, replacing constant-density approximations. **CORSIKA** (Heck et al., 1998) is used in an exploratory capacity to generate primary cosmic-ray samples and shower-level neutrino yields.

3 Methodology

The methodology followed in this work has five parts. section 3.1 establishes that the translated core is numerically faithful to **PEANUTS** and computationally competitive, the precondition for the rest of the chapter. sections 3.2 and 3.3 describe how that validated core is extended into a modular propagation framework and into beyond-the-Standard-Model scenarios. section 3.4 then closes the loop by validating the full source–medium–detector–inference chain against real data, and section 3.5 documents the release of the resulting code as a public, reusable package for the community.

3.1 Core translation: methodology, validation and benchmarking

3.1.1 Translation methodology

The development of `tpeanuts` follows a systematic translation process: each PEANUTS (Gonzalo & Lucente, 2024) module (PMNS construction, vacuum and matter Hamiltonians, matter mixing, numerical propagation, solar production and Earth propagation, detector exposure) is first interpreted from the underlying theoretical formalism of section 2.1, separating the physical algorithm from implementation-specific detail, and then reformulated using native, batched PyTorch (Paszke et al., 2019) tensor operations, so that large grids of energies, baselines and production points are evaluated within a single vectorised call. PEANUTS’s physical conventions, the PMNS phase, the neutrino/antineutrino sign, mass-eigenstate ordering, matter-potential normalisation, are explicitly verified at every step: intermediate quantities (mixing matrices, Hamiltonians, eigenvalues, matter potentials, oscillation phases) are compared directly against the original implementation before validating the final oscillation probabilities, since sign or unit-conversion errors can otherwise produce numerically stable but physically incorrect results.

Central to this translation is the batched PyTorch tensor formulation detailed architecturally in section 4.3: every quantity, from PMNS matrices to full transition probabilities, carries an explicit batch dimension, and every operation on it is applied simultaneously across that dimension by ordinary broadcasting, so that a parameter scan, a detector-response folding over thousands of energy bins, or a gradient-based fit all reuse the same vectorised call rather than looping explicitly in Python, moving between CPU/GPU and single/double precision with no change to the physics code. The low-level physics primitives preserve the autograd graph by default; the fast-forward routines behind most results in sections 5.3 and 5.4 wrap this in `torch.no_grad()` for speed, while the inference models of section 3.4 build on the same differentiable primitives directly, so the same gradient constructing the oscillation Hamiltonian also drives the LBFGS fit and the Laplace/Fisher uncertainty estimate.

3.1.2 Validation strategy

Validation proceeds in two independent directions. **Intrinsically**, the perturbative/analytical and the purely numerical (matrix-exponential) evolvers of section 2.1 solve the same propagation problem by independent strategies and are checked against each other, to a common tolerance, wherever both apply. **Extrinsically**, numerical, code-to-code agreement is verified hierarchically against two independent references:

- low-level quantities, such as PMNS matrix elements, kinetic phases and matter potentials;
- intermediate physical outputs, such as effective mixing angles and eigenvalues in matter and mass-eigenstate weights for solar neutrinos;
- final oscillation probabilities, energy- and production-averaged and, where a detector model exists, folded into predicted event rates.

Since PEANUTS only implements solar-neutrino propagation, the comparison against it is restricted to that regime: vacuum propagation, the solar MSW effect, production-weighted survival probabilities, and Earth regeneration. `nuSQuIDS` (Argüelles et al., 2022), which solves the propagation problem independently for both the solar and the atmospheric regimes, cross-checks

the full scope of the implementation instead and, relying on a different numerical method, also exposes discrepancies from conventions shared between PEANUTS and `tpeanuts` but absent from an independent solver. Agreement is verified to machine precision for vacuum and atmospheric propagation, where `tpeanuts` and the reference codes solve the same closed problem; for solar and solar-Earth-detection observables, the comparison against `nuSQuIDS` instead shows percent-to-ten-percent-level differences, attributable to genuine convention and approximation differences rather than floating-point or integration error. Probability conservation, positivity, and unitarity of the evolution operator are checked throughout.

3.1.3 Benchmarking

Beyond physical correctness, an important objective is to evaluate computational performance, both **intrinsically** (CPU vs GPU, float32 vs float64, numerical versus perturbative, etc.) and **extrinsically** (against PEANUTS and `nuSQuIDS`), since efficient evaluation of large numbers of configurations is a principal motivation for the tensor-based design.

Intrinsic benchmarking. Executing the batched evaluation of section 3.1 on a GPU means thousands of arithmetic units working through the whole batch in parallel, rather than one core working sequentially through a list: launching a computation has a fixed cost independent of batch size, so small batches favour a sequential implementation while large batches amortise that cost and let GPU parallelism dominate, the key to reading the results of section 5.1.2. Tensors are pre-allocated on the target device, GPU timings are taken only after explicit synchronisation, and warm-up iterations are excluded from the reported, repetition-averaged timings; both CPU and GPU are evaluated in single- and double-precision arithmetic, the latter for direct comparison with the reference codes.

Extrinsic benchmarking. The same oscillation parameters, density profiles, precision and scenarios are used across all three packages wherever the underlying calculation is directly comparable, vacuum and atmospheric propagation in particular. The suite varies the number of energy points, baselines/density samples, batch size, device, precision and oscillation scenario; the metric reported throughout this thesis is wall-clock speed-up relative to PEANUTS and `nuSQuIDS`. `nuSQuIDS` evaluates one solver object per energy/angle point, while PEANUTS already vectorises with NumPy over a single trajectory; `tpeanuts` batches across the full grid at once, so its relative advantage over `nuSQuIDS` grows steadily with grid size, while over PEANUTS it only emerges once the batch is large enough to amortise the fixed cost of dispatching PyTorch operations.

3.2 Extension to a modular architecture: sources, media and detectors

The validated core of section 3.1.2 is deliberately decoupled from any specific neutrino source, propagation medium or detector: the same Hamiltonian and evolution machinery serve solar, atmospheric and reactor sources; vacuum, solar and Earth matter media, with interchangeable density models; and the detector models of table 2 built on top of the resulting probabilities. The same engine could serve accelerator sources with no change to the propagation core. This separation is what allows the framework to extend beyond PEANUTS’s original solar-only scope, to atmospheric-neutrino propagation through the Earth, alternative solar and Earth density models, and new detectors, without touching the propagation core already validated above.

3.3 BSM extensions: NSI and sterile neutrinos

Beyond-the-Standard-Model scenarios are incorporated as modifications of the same Hamiltonian (sections 2.3.1 and 2.3.2), validated by a signature specific to each extension. For **NSI**, the benchmark is the LMA-Dark degeneracy (Esteban et al., 2018; Miranda et al., 2006): the simultaneous transformation $\theta_{12} \rightarrow \pi/2 - \theta_{12}$, $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$ must reproduce a day–night–asymmetry sign flip consistent with the near-degenerate solution, exercising the sign dependence of the implementation. For **sterile neutrinos**, three presets (a null-mixing reference, a global-fit-motivated best fit, and a short-baseline-anomaly benchmark) are checked against analytic expectations: exact recovery of the Standard-Model limit at zero sterile mixing and unitarity of the enlarged 4×4 mixing matrix to machine precision..

3.4 Inference methodology: the full source–medium–detector chain

Validating the detector and inference layers exercises the full chain built in the sections above end to end, source flux, medium propagation, detector response and statistical inference, against real, published experimental data, since no independent reference implementation exists to compare against. Three complementary checks are applied, in full where the detector model is complete and in part otherwise, across the implemented experiments:

- a **closure test**, a prediction-reproduction check against real published data, evaluating the full forward model at the experiment’s own published best-fit parameters.
- an **absolute-normalisation audit**, when the closure test reveals a systematic offset, tracing it through every stage of the forward-model chain against independently published values to identify and quantify its dominant sources;
- a **parameter-recovery fit**, minimising the appropriate likelihood of section 2.4 with the quasi-Newton LBFGS optimiser (Liu & Nocedal, 1989), comparing the fitted values against the experiment’s own published best fit and against the global-fit value from NuFIT 6.1 (Esteban et al., 2024, 2025).

This methodology is applied in full to Daya Bay, the primary case study of section 5.5.1; in summary form, with a parameter-recovery fit, to IceCube DeepCore, SNO phase I and SNO phase II; and only up to the implementation/unit-test stage, with no fit to real data yet reported, for Borexino (section 5.5.2).

3.5 Open-source release: a public, documented software package

Releasing **tpeanuts** as public, open-source software was a deliberate part of this methodology of work. The full source code, together with the notebooks used for every validation, benchmark and analysis in this thesis, is publicly available on GitHub under the GNU General Public License v3 (GPLv3), the same licence as the original PEANUTS implementation it builds on, with an editable, development-mode install (`pip install -e .`) and a documented layout. Beyond the code itself, six companion L^AT_EX documents provide module-by-module architectural and physics detail beyond the scope of this thesis, each with its own figures and, for the inference documents, its own real-data results. Full details, including the repository URL and the notebook directory tree, are given in section 5.6.

4 Technical Development

`tpeanuts` takes PEANUTS as its starting point, but structures the resulting code in a modular, hierarchical way that favours extending it with new features and physical models, rather than reproducing PEANUTS's own flat structure. This section describes the main components of that structure.

4.1 Software architecture

`tpeanuts` is a PyTorch-native reimplementation of the neutrino oscillation and flux-propagation toolkit originally developed as PEANUTS (a NumPy/Numba package), extended into a full detector-level statistical-inference framework. The guiding design principle is to replace the scalar, per-point evaluation model of the original implementation with a batched, tensor-oriented formulation: every physical quantity, Hamiltonians, evolution operators, oscillation probabilities, fluxes, predicted event counts, likelihoods, is expressed as a function of broadcastable tensors of Torch, so that large grids of energies, angles and detector configurations, within memory limits, can be evaluated in a single vectorised call.

The underlying physics primitives also support automatic differentiation, or autograd: Torch's mechanism for recording every tensor operation applied during a computation and replaying that record to obtain the exact derivative of its output with respect to any of its inputs, without deriving or coding those derivatives by hand. The fast-forward simulation routines used for most physics results disable autograd in Torch for speed, while the experiment-specific inference models of section 4.1.1 build on the same primitives with autograd enabled, so that gradients can be backpropagated through the full chain for the gradient-based fits of section 3.4. This is the central architectural departure from both the legacy implementation and from reference tools such as `nuSQuIDS` (Arguelles et al., 2022), whose C++/pybind11 API is fundamentally scalar, with one solver object per energy/angle point.

`tpeanuts` therefore operates in two complementary modes:

- **Fast-forward simulation:** dedicated pipelines assemble the propagation primitives into complete, physically meaningful observables for each production environment, a solar-neutrino pipeline and an atmospheric-neutrino pipeline being the two main instances, evaluated with autograd disabled for speed.
- **Inference:** the same primitives are reused with autograd enabled, so that gradients propagate through the full chain, from the free oscillation parameters down to the predicted observable, letting a gradient-based optimiser fit those parameters directly against real, published experimental results.

Figure 1 sketches the resulting three-tier decomposition, application, domain and core layers, detailed below. This hierarchy is itself supported by a set of blocks that sit outside it: a configuration block centralising model choices, a shared utility layer providing standard device/precision management and generic numerical primitives, and a collection of external-tool integrations, third-party packages such as `MCEq` and external data repositories such as the Honda/HKKM flux tables, consumed as configuration input to the physics above.

Concretely, this three-tier architecture is implemented as a set of independent Python packages,

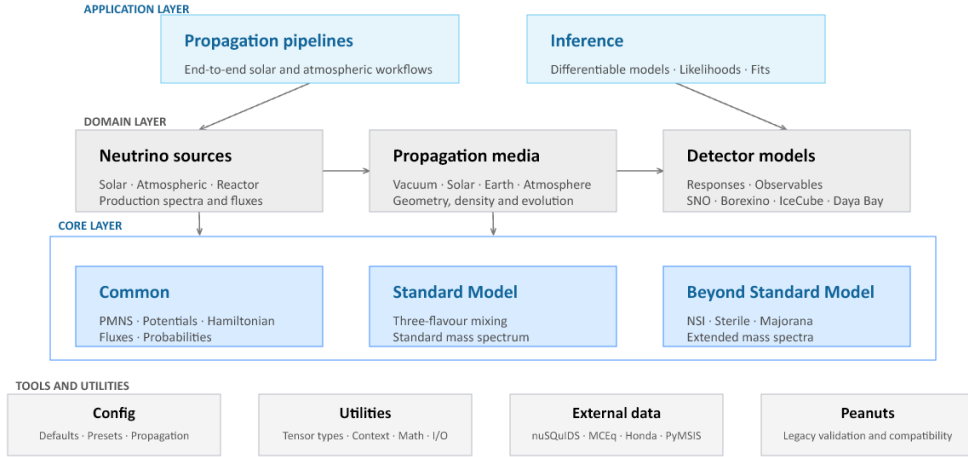


Figure 1: Modular architecture of **tpeanuts**: an application layer (propagation pipelines, inference) built on a domain layer (neutrino sources, propagation media, detector models), which specialises a shared core (common primitives, Standard Model and BSM Hamiltonians) and is supported by configuration, utility, external-data and legacy-PEANUTS-compatibility tools.

one per node of figure 1, summarised in table 1 together with the responsibility of each.

Table 1: Packages composing **tpeanuts**, and their responsibility.

Package	Responsibility
config/	Composed configuration objects and named SM/BSM/NSI presets, replacing per-call keyword arguments.
external/	Optional third-party integrations (MCEq, nuSQuIDS, Honda, PyMSIS) for cross-validation and flux/density inputs.
util/	Device/dtype management (<code>RuntimeContext</code>) and generic numerical primitives.
core/	Medium-independent Hamiltonians, PMNS/mass-spectrum construction, evolution operators; standard and BSM (3 + 1 sterile, NSI, Majorana) variants.
source/	Emission models at the neutrino source: solar production profiles, reactor (Huber–Mueller) antineutrino spectra.
medium/	Environment-specific density/geometry models: vacuum, solar, Earth, atmosphere.
detector/	Forward-folding of oscillated flux into predicted event counts: cross sections, target composition, energy response, efficiency, backgrounds, per experiment.
pipeline/	Composition of the above into complete observables (solar/Earth/atmosphere detection chains).
inference/	Likelihood construction, LBFGS gradient-based fitting, Laplace/Fisher uncertainty estimation, likelihood scans.

4.1.1 Core, Domain and Application layers

Core Layer. At the base of the hierarchy, the core layer implements the medium-independent building blocks shared by every propagation scenario, for the plain three-flavour Standard Model and for the optional BSM extensions (3 + 1 sterile, NSI), together with an optional Majorana mass-matrix parametrisation relevant to lepton-number-violating observables rather than standard oscillation probabilities: the vacuum and matter Hamiltonians, PMNS/mass-spectrum construction, the matter potential, and the machinery to convert a Hamiltonian into a unitary evolution operator.

Two complementary propagation strategies are implemented here:

- **perturbative/analytical** evolver for smooth, slowly varying density profiles.
- **numerical** evolver, medium-independent, which discretises the trajectory, exponentiates

a local Hamiltonian on each segment, and composes the segment operators through a logarithmic-depth binary-tree reduction, keeping the composition fully batchable and GPU-friendly.

Domain Layer. The domain layer specialises this shared core for each physical scenario, adding the neutrino-source, propagation-medium and detector models that the core layer itself knows nothing about. A source model isolates the physics of neutrino *emission*, upstream of propagation: solar production distributions from the Standard Solar Model, atmospheric neutrino profile production from MCEq/Honda, and the per-isotope Huber–Mueller (Huber, 2011; Mueller et al., 2011) reactor-antineutrino emission spectrum. A propagation-medium model then specialises the core with an environment-specific density/geometry treatment: vacuum, solar (adiabatic MSW/Landau–Zener mixing with production-point-averaged fluxes), Earth (matter regeneration along nadir-angle trajectories, with exposure time-averaged over the detector’s latitude and the Earth’s rotation), and atmosphere (propagation from a production height through the atmosphere).

Finally, a detector model folds an oscillated flux into a predicted, binned event count observable, entirely decoupled from how the probability was computed: a target composition, an energy response, an efficiency and a background, common to every experiment, combined with the relevant cross section by physical process: inverse beta decay to order $1/M$ (Vogel & Beacom, 1999) (reactor antineutrinos), neutrino–electron elastic scattering (solar), and deuteron dissociation (SNO). Five experiment-specific modules specialise this for Borexino, Daya Bay, IceCube DeepCore, and SNO phases I and II, summarised in table 2.

Application Layer. At the top of the hierarchy, the application layer composes the models of the domain layer into complete, physically meaningful results. On one side, propagation pipelines assemble the source, medium and detector models into end-to-end observables, a solar-production-to-Earth-detection chain, or a full atmospheric flux propagated through the Earth.

On the other side, built on the same validated primitives, the statistical-inference machinery implements the formalism of section 2.4 as reusable, detector-agnostic components: a common differentiable-model interface, a free-parameter tuple and a prediction function, that every concrete model, oscillation-only or detector-composed, implements, together with:

- a set of **likelihood functions** covering Poisson counts, asymmetric Gaussian ratios and, for covariance-matrix observables such as SNO II’s 38-observable data set, a correlated Gaussian likelihood, together with a Gaussian penalty term for externally constrained nuisance parameters;
- an **LBFGS minimiser** paired with the Laplace/Fisher covariance estimate;
- a **two-dimensional likelihood-scan** routine, with an optional per-point re-minimisation of the remaining free parameters.

The concrete, experiment-specific models composing an oscillation model with a detector’s event-rate function live alongside each detector.

4.1.2 External-tool integrations (external/)

This layer isolates optional third-party dependencies from the core physics code:

- `external/mceq/` wraps the MCEq cascade-equation solver for atmospheric neutrinos (Fedynitch et al., 2015);
- `external/nusquids/` wraps nuSQuIDS (Arguelles et al., 2022) as an independent reference for validation and benchmarking;
- `external/honda/` reads the Honda/HKKM atmospheric-flux tables (Honda et al., 2007);
- `external/pymis/` provides an empirically calibrated atmospheric density model (Emmert et al., 2021).

Each is imported lazily so the package degrades gracefully when the optional extras are not installed.

Table 2: Experiments implemented on the detector/inference layers, and their validation status against real published data (section 5.5.2).

Experiment	Channel / detector physics	Status against real data
Daya Bay	Reactor $\bar{\nu}_e$ IBD, 8 detectors, 6 reactors, full flux/response chain	Near/far fit recovers $\sin^2 2\theta_{13}$, Δm_{31}^2 jointly within 1–2% of published (section 5.5.1); full-shape fit not yet robust
IceCube DeepCore	Atmospheric ν_μ disappearance, event-by-event weighted-MC reweighting	Fit recovers θ_{23} , Δm_{3l}^2 within -0.05σ of published; fitted muon-background normalisation traced to fixed detector systematics
SNO II (salt phase)	CC/NC/ES, 38 published observables, published statistical covariance plus a reconstructed systematic covariance	Fit converges to θ_{12} , Δm_{21}^2 , λ_{8B} consistent with SNO’s own SNO-only range
Borexino	Solar ν - e elastic scattering, pp-chain spectrum	Detector response and event-rate chain implemented; background model and energy-resolution normalisation not yet taken from the published analysis
SNO (phase I)	CC/NC/ES, pure-D ₂ O phase	Day/night spectral fit to real published data recovers $\theta_{12} = 34.33^\circ \pm 0.67^\circ$, 0.86σ from NuFIT 6.1 (section 5.5.2)

4.1.3 Notebooks: validation, benchmarking, analysis, reproduction and inference

Beyond the architecture described above, a curated suite of notebooks exercises every layer of `tpeanuts` and constitutes the primary, reproducible evidence base behind every result reported in this thesis. The suite is organised into five complementary groups.

- **Validation and benchmark** notebooks establish bit-level agreement against legacy PEANUTS and cross-validate against `nuSQuIDS` for every propagation medium, and measure CPU/GPU performance against both references.
- **Solar and atmospheric analysis** notebooks explore the fundamentals of each propagation regime, production and exposure profiles, differential-flux structure, and the behaviour of the underlying density models, independently of any specific detector or experiment.
- **Experiment Reproduction** notebooks recover well-known physical signatures, the solar day–night asymmetry, the MSW resonance, atmospheric oscillograms, Earth matter tomography, and the LMA-Dark degeneracy, establishing that the implementation reproduces established physics independently of any single reference code.

- **BSM extensions** notebooks validate the non-standard-interactions and sterile-neutrino extensions independently of the Standard-Model reproduction above: NSI notebooks compare the solar ^8B survival probability under literature NSI presets against the Standard-Model limit and scan individual off-diagonal couplings; sterile-neutrino notebooks confirm the $3 + 1$ extension's Standard-Model limit, unitarity and vanishing sterile matter coupling under the CC-only default, together with its expected CP-violation and short-baseline sensitivity.
- **Inference** notebooks close the chain, running real-data prediction checks, normalisation audits and parameter-recovery fits where the experimental model is sufficiently complete (Daya Bay, IceCube DeepCore, SNO phase I and SNO phase II); Borexino is currently validated only at component and unit-test level.

4.2 Numerical simulation conventions

The numerical core of `tpeanuts` is built around a small number of deliberate conventions that keep every propagation medium numerically consistent, dimensionally well-behaved, and directly comparable to both the legacy PEANUTS implementation and to `nuSQuIDS`. The most important of these are the choice of a unitary, dimensionless evolution scale for the Hamiltonian, the neutrino/antineutrino sign convention, and the floating-point precision policy used throughout the matrix-exponential evolution.

4.2.1 Non-dimensionalisation of the Hamiltonian

Rather than working directly with the physical matter potential and kinetic (mass-squared) terms, which span many orders of magnitude depending on the neutrino energy and the medium's density, every Hamiltonian contribution is first converted into a dimensionless quantity by multiplying it by a configurable evolution length scale, `evolution_scale_m` (the Earth radius, by default). Concretely,

$$V = s L_{\text{scale}} \sqrt{2} G_F N_A (10^6 \text{ cm}^3/\text{m}^3) (\hbar c)^2 n_e, \quad k_i = L_{\text{scale}} \frac{\Delta m_i^2}{2E\hbar c}, \quad (18)$$

where $s = \pm 1$ is the neutrino/antineutrino sign (see below), N_A is Avogadro's number and the $10^6 \text{ cm}^3/\text{m}^3$ factor converts n_e , the electron number density in mol/cm^3 , to mol/m^3 consistently with L_{scale} in metres, and Δm_i^2 are the (reduced) mass-squared eigenvalues. The trajectory coordinate is normalised by the same scale, $x = L/L_{\text{scale}}$, so that the evolution operator is always computed as $U = \exp(-iHx)$, with both H and x dimensionless.

This convention has two practical benefits for a batched, GPU-oriented implementation: it gives a uniform dimensionless convention for every propagation medium, avoiding mixed physical length units inside the evolutors, without changing the physical phase $\hat{H}x = H_{\text{phys}}L$ or the matrix argument passed to the exponential; and it decouples the physical unit system from the propagation length, since only the ratio L/L_{scale} ever enters the exponent. The evolution scale defaults to the Earth's radius for backward compatibility with the legacy PEANUTS convention, but is an explicit, per-medium parameter, settable e.g. to the injection height of an atmospheric shower rather than to $R_{\oplus}$.

4.2.2 Neutrino/antineutrino convention

Antineutrino propagation is selected through a single boolean (or boolean-tensor, for batched mixed samples) flag, `antinu`, carried by the `OscillationParameters` object and threaded through every Hamiltonian and probability routine. Two independent sign flips are applied consistently whenever `antinu=True`:

- **Matter potential.** The Wolfenstein charged-current potential changes sign, $V \rightarrow -V$, since it originates from the coupling of ν_e (not $\bar{\nu}_e$) to the ambient electrons.
- **Mixing matrix.** The PMNS matrix is replaced by its complex conjugate, $U \rightarrow U^*$, equivalent to flipping the sign of the CP-violating phase, $\delta_{\text{CP}} \rightarrow -\delta_{\text{CP}}$.

Both substitutions are applied at the point where the reduced mixing matrix and the matter Hamiltonian are assembled, so that every downstream propagation routine, vacuum, solar, Earth or atmospheric, inherits the correct antineutrino behaviour automatically from a single flag, without medium-specific antineutrino logic. The same object also encodes the neutrino mass ordering as the sign of the atmospheric splitting Δm_{3l}^2 : a positive value selects normal ordering and a negative value selects inverted ordering, so that mixed-ordering batches can, in principle, be evaluated in one call.

4.2.3 Precision and evaluation mode

All propagation is carried out at double precision (`torch.float64` real / `torch.complex128` complex) by default, matching the precision of the legacy NumPy implementation and keeping the matrix-exponential composition numerically stable over trajectories segmented into hundreds of steps. The fast-forward simulation pipelines are wrapped in `@torch.no_grad()` by default, since most forward evaluation, benchmarking and physics reproduction does not need gradients and disabling autograd bookkeeping reduces memory footprint and computational overhead; the underlying physics primitives themselves, and the experiment-specific inference models built directly on them (section 4.1.1), do not disable autograd, and it is this differentiable path, not the fast pipelines, that drives the gradient-based fitting of section 3.4.

For validation purposes, a `legacy_precision` flag is threaded through the potential and Hamiltonian builders; when enabled, it substitutes the full-precision matter-potential prefactor, derived from CODATA G_F , N_A and $\hbar c$ (Tiesinga et al., 2021), with the four-significant-digit constant hard-coded in the original PEANUTS implementation, allowing bit-level, rather than merely physics-level, agreement to be verified against the legacy reference where required.

4.3 Neutrino detection pipelines

Beyond the medium-specific propagation primitives described above, the framework assembles them into two complete, physically distinct detection workflows: a solar-neutrino pipeline, which propagates neutrinos produced deep inside the Sun to a terrestrial detector, and an atmospheric-neutrino pipeline, which propagates neutrinos produced by cosmic-ray air showers in the Earth’s atmosphere down to a detector. Both share the same underlying core Hamiltonian/evolutor machinery, but differ fundamentally in one respect: whether the mass-eigenstate coherence acquired at production survives all the way to the detector.

4.3.1 Solar-neutrino detection pipeline

A solar neutrino is produced as a pure ν_e at some radius ρ inside the Sun, distributed according to a source-specific production profile (e.g. ${}^8\text{B}$, pp) (Vinyoles et al., 2017). As it propagates outward through the Sun’s smoothly varying electron density, the MSW matter effect adiabatically rotates the flavour state into the local mass-eigenstate basis, with a small non-adiabatic level-crossing correction (Parke, 1986). Because the $\sim 1.5 \times 10^8$ km vacuum baseline to Earth is enormously larger than the oscillation length at solar-neutrino energies, the relative phases between mass eigenstates wash out completely before arrival: only the incoherent mass-eigenstate populations survive.

`tpeanuts` implements this picture as a pipeline reproducing, in batched `torch` form, the original PEANUTS NumPy/Numba reference workflow used to validate it bit-for-bit. The pipeline proceeds in three stages:

1. **Solar production and adiabatic conversion:** for a given source and energy grid, the electron-neutrino state is propagated adiabatically outward from each production radius ρ in the source’s normalised production-radius distribution, and the result is averaged over that distribution to yield a single vector of mass-eigenstate weights, $w_i(E) = (w_1, w_2, w_3)$, the fraction of the flux exiting the Sun in each mass eigenstate.
2. **Vacuum propagation to Earth:** because no relative phase is ever tracked, propagation across the Sun–Earth vacuum baseline leaves the mass weights $w_i(E)$ unchanged by construction; this step is a physical no-op in the incoherent formulation, not an approximation applied afterwards.
3. **Earth matter propagation:** at the detector, the incoherent mixture of incident mass eigenstates is propagated through Earth matter, relevant only for underground detectors sensitive to the day/night matter-regeneration asymmetry, evaluated over a grid of nadir angles η ; each mass eigenstate acquires its own Earth-matter transition probability to each final flavour α , $P_{i \rightarrow \alpha}^\oplus(E, \eta)$, and the final flavour probability is their incoherent sum, $P_{\nu_e \rightarrow \nu_\alpha}(E, \eta) = \sum_i w_i(E) P_{i \rightarrow \alpha}^\oplus(E, \eta)$.

This oscillation probability, still a function of energy rather than a detector observable, is further averaged: over the solar production-radius distribution (folded into the first stage), over the neutrino energy spectrum, and, for underground detectors, over the detector’s nadir-angle exposure, the fraction of a year during which each nadir angle is observed, computed from the detector’s geographic latitude and the Earth’s daily and annual motion; converting it into a predicted event rate additionally requires the flux, cross section and detector response.

4.3.2 Atmospheric-neutrino detection pipeline

Atmospheric neutrinos follow a different physical picture. They are produced not at a single well-defined point but across a broad, energy- and angle-dependent distribution of production heights h in the atmosphere, with the boundary-condition flux $\Phi_\beta(E, h, \theta)$ for each produced flavour β supplied by an external cascade-equation generator (MCEq Fedynitch et al., 2015) or by tabulated Honda/HKKM fluxes (Honda et al., 2007), which provide ν_e , $\bar{\nu}_e$, ν_μ and $\bar{\nu}_\mu$; the negligible conventional ν_τ flux is carried with zero initial value. Unlike the solar case, coherence is not lost here: even for an upward-going, Earth-crossing trajectory of up to $\sim 1.3 \times 10^4$ km

(section 2.1), the mass-eigenstate wave packets remain overlapped at these energies and detector resolutions, so the full flavour-amplitude state is propagated coherently from production to detector. The pipeline proceeds in five stages:

1. **Flux selection:** one produced particle and one detector zenith angle are sliced out of the loaded height/energy/angle flux dataset.
2. **Atmosphere propagation:** a coherent unit-amplitude flavour state $|\beta\rangle$ is evolved from each production height down to the Earth’s surface, batched jointly over the full energy $\times$ height grid.
3. **Earth propagation:** for an upward-going, Earth-crossing trajectory, the surface state is propagated further through Earth matter to the detector depth, using the same coherent evolver as the solar pipeline’s coherent variant.
4. **Height integration:** the detector-level flux for each final flavour α is obtained by weighting the propagated $\beta \rightarrow \alpha$ probability at each height by $\Phi_\beta(E, h, \theta)$ and trapezoidally integrating over h .
5. **Flavour summation:** the height-integrated contributions from every produced flavour (ν_e, ν_μ, ν_τ and antiparticles) are summed into the total detector-level flux of each final flavour.

4.4 Test suite and sanity checks

Correctness in `tpeanuts` is established through a dedicated automated test suite of 942 test functions organised into 64 files that mirror the package’s module tree one-to-one. The suite is executed manually with `pytest` after relevant modifications, and falls into four broad, partly overlapping categories, complemented by a further, notebook-level layer of validation:

- **Code consistency checks** verify the software contract of each function independently of the underlying physics: output shape/dtype/device, broadcasting over batched grids, exceptions on invalid inputs, consistency between alternative code paths (e.g. “analytical” vs. “numerical”), and save/load round-trips. This is the largest category, roughly a third to two-fifths of the suite.
- **Mathematical checks** verify that a function’s output matches an independently derived closed-form expression or exact algebraic identity: the PMNS matrix’s unit-modulus determinant and Jarlskog invariant; the composed segment-evolver product against a manually accumulated matrix product; agreement between the perturbative and purely numerical evolvers on a constant-density segment; a resonance radius against its closed-form solution; and every non-dimensionalised term against its defining formula under parameter scaling. Roughly a quarter to a third of the suite.
- **Physical coherence checks** verify that computed quantities respect the physical constraints of the theory, independently of how they were computed: probability conservation (every complete, N -flavour unitary-evolution transition matrix doubly stochastic); unitarity of every evolution operator; recovery of known physical limits (zero baseline/density reduces to the identity, the adiabatic solar limit as the Landau–Zener probability vanishes, the expected antineutrino day/night asymmetry direction); and, most stringently, function-

by-function cross-validation against the original PEANUTS NumPy/Numba reference. Roughly a third to two-fifths of the suite.

- **Detector and inference validation:** because these layers have no independent reference implementation to compare against, their unit-test coverage focuses on code-consistency and mathematical checks instead, e.g. the invertibility of the Daya Bay experimental model’s $(\theta_{13}, \Delta m_{31}^2) \leftrightarrow (\sin^2 2\theta_{13}, \Delta m_{ee}^2)$ reparametrisation on its restricted physical branch, or that the $\text{IAV} \rightarrow \text{LSNL} \rightarrow \text{Gaussian}$ -resolution response composition conserves total probability; the physical-coherence role played elsewhere by comparison with PEANUTS/nuSQuIDS is instead played by the closure tests and parameter-recovery fits of section 3.4, run as notebooks rather than automated test cases.
- **Notebook-level validation:** beyond the automated test suite, a second, complementary layer of validation lives in a dedicated set of notebooks, which reproduce bit-level agreement against legacy PEANUTS via `legacy_precision` and cross-check against nuSQuIDS (Argüelles et al., 2022) within scenario-dependent tolerances, machine precision for vacuum and atmospheric propagation, percent-to-ten-percent-level for solar and solar-Earth-detection observables (section 3.1.2), for every propagation medium; run real-data prediction checks and, where the detector model is sufficiently complete, the normalisation audits and parameter-recovery fits of section 3.4; and embed inline “smoke test” cells, printed diagnostics followed by assertion statements, that must pass before the notebook is considered valid.

5 Validation and Results

This section presents the validation outputs, benchmark results, simulation analyses and phenomenological applications obtained for `tpeanuts`, following the methodology described in section 3. Validation remains essential throughout, since it establishes the reliability the rest of the chapter builds on, but from section 5.2 onward the results also demonstrate new capabilities of the framework, a distinction kept explicit below, together with the level of maturity of each result: fully validated, implemented, or exploratory.

Section 5.1 establishes that the translated implementation is numerically reliable against PEANUTS and nuSQuIDS, and computationally useful, through the CPU/GPU benchmarks whose overhead-versus-parallelism behaviour is explained in section 3.1.3. Section 5.2 then demonstrates capabilities that go beyond a translation of PEANUTS: vacuum, solar and atmospheric-neutrino propagation through the Earth, and flexible Earth/matter models. Section 5.3 builds on that same engine to reproduce a range of non-trivial, well-known physical phenomena. Section 5.4 extends this to non-standard interactions and sterile neutrinos. Section 5.5 shows that the framework can be used for phenomenology and inference with real experimental data. Finally, Section 5.6 documents the public, reusable release of the code underlying every result above.

5.1 Validation and benchmark

5.1.1 Internal validation

Beyond comparisons against external reference codes, the perturbative/analytical and the purely numerical (`matrix_exp`-based) evolutors of section 2.1 solve the same propagation problem through independent numerical strategies and must therefore agree with each other; the same self-consistency logic also applies to the discrete implementation choices within each evolver, reunitarizing the truncated perturbative operator or not, the closed-form (Cardano) versus dense eigenvalue solver, the Landau–Zener correction, and the runtime `RuntimeContext` (float32/float64 precision, CPU/CUDA device), all of which should leave the physical result unchanged to within the tolerance of the corresponding approximation. Table 3 summarises these checks for the Standard Model.

Table 3: Standard Model intrinsic self-consistency: maximum and mean absolute and thresholded-relative errors.

Comparison	max abs	mean abs	max rel	mean rel
Solar: <code>adiabatic_approximated</code> vs <code>numerical</code>	4.88×10^{-3}	3.85×10^{-4}	4.58×10^{-2}	2.50×10^{-3}
Solar: <code>adiabatic_exact</code> vs <code>numerical</code>	4.87×10^{-3}	3.80×10^{-4}	4.56×10^{-2}	2.47×10^{-3}
Earth: perturbative vs numerical	3.22×10^{-3}	1.32×10^{-4}	9.24×10^{-3}	3.96×10^{-4}
Atmosphere: perturbative vs numerical	1.44×10^{-11}	1.86×10^{-13}	1.07×10^{-8}	1.76×10^{-11}
Reunitarize off vs on:	1.50×10^{-2}	1.01×10^{-3}	2.05×10^{-2}	3.26×10^{-3}
Eigenv.: Cardano vs <code>torch.linalg.eigvalsh</code>	9.06×10^{-13}	1.56×10^{-14}	1.63×10^{-11}	1.19×10^{-13}
Landau–Zener on vs off (solar surface)	0	0	0	0
Precision: numerical float32 vs float64	1.04×10^{-4}	1.11×10^{-5}	1.75×10^{-4}	3.35×10^{-5}
Device: numerical CPU vs CUDA (float64)	4.92×10^{-14}	2.38×10^{-15}	3.06×10^{-13}	1.02×10^{-14}

The adiabatic solar-propagation methods, in both their exact and approximated forms, differ from the numerical reference by absolute errors of order 10^{-3} in the computed transition probabilities, confirming the validity of the adiabatic approximation for the NuFIT 5.2 preset used throughout this thesis. Enabling the Landau–Zener correction produces no appreciable difference with respect to leaving it disabled, a further confirmation of the accuracy of the adiabatic approximation at this preset. The perturbative evolver differs from the purely numerical one also by errors of order 10^{-3} for Earth propagation, confirming its validity at that precision level; reunitarizing the truncated perturbative Hamiltonian removes discrepancies of order 10^{-2} , appreciably improving the agreement. In the atmosphere the perturbative-numerical discrepancy never exceeds 10^{-8} , fully validating the perturbative method there. The closed-form eigenvalue computation, Cardano’s expressions for the three-flavour Hamiltonian and, subsequently, Ferrari’s for the four-flavour Hamiltonian used in the BSM sterile extension, agrees with the diagonalization obtained through `torch`’s standard tools at essentially bit-level precision. Regarding the runtime context, moving the computation to GPU/CUDA does not affect precision at all, whereas using float32 can introduce discrepancies of up to 10^{-4} . The overall conclusion is that the fast processing path, adiabatic approximations in the Sun combined with perturbative methods in Earth and the atmosphere and using the reunitarize option, introduces inaccuracies of order 10^{-3} , so that running this path in float32 on CPU or CUDA can appreciably improve speed and save processing resources and time.

5.1.2 PEANUTS validation and performance comparison

Bit-level agreement with the legacy PEANUTS implementation is the primary correctness criterion for the translation. Figure 2 (left) summarises the maximum absolute and relative differences observed across the four validated modules, solar, Earth, vacuum and the composed detection pipeline, when both codes are evaluated with the `legacy_precision` flag enabled on matched B16-AGSS09 input tables for the solar-dependent modules. All four modules now agree at or below the 1 ppm reference line: solar agrees to $\sim 5.5 \times 10^{-15}$, Earth to $\sim 5.4 \times 10^{-10}$ and the composed pipeline to $\sim 2.6 \times 10^{-9}$, all three below the 1 ppb reference line, and vacuum agrees to $\sim 1.5 \times 10^{-7}$; every module is consistent with the bit-level-agreement expectation this comparison is designed to test.

Figure 2 (right) shows the wall-clock speed-up of `tpeanuts` over legacy PEANUTS for the Earth-propagation probability calculation, as a function of the energy-grid and nadir-grid sizes used in the batched evaluation. For small grids `tpeanuts` is in fact slower than PEANUTS, by up to a factor ~ 10 at the smallest grid considered (4×4), and only becomes faster, on average, once the grid exceeds a few tens of points per dimension, reaching a peak of $\sim 5.7\times$ around grid sizes of several hundred points per dimension (e.g. 512×512), with a slight decrease at the single largest grid considered (1024×1024 , $\sim 4.9\times$). This is the expected outcome of comparing against an already vectorised NumPy implementation rather than against a scalar, per-point solver: PEANUTS batches efficiently over array operations for a single trajectory, so the advantage of `tpeanuts`’s tensor-based formulation only emerges once the batch size is large enough for the fixed overhead of constructing and dispatching PyTorch operations to be amortised, consistent with the benchmarking methodology of section 3.1.3.

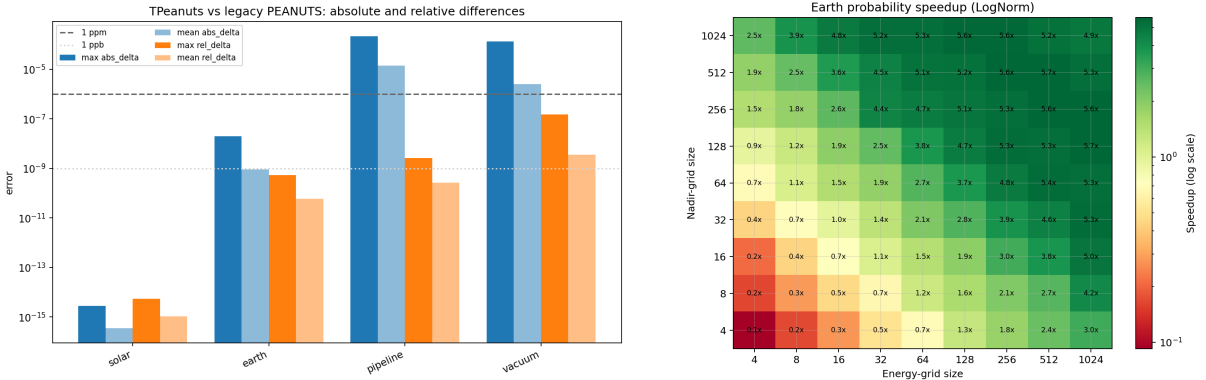


Figure 2: PEANUTS validation and performance: maximum and mean absolute and relative differences between `tpeanuts` and legacy PEANUTS, by validated module, dashed and dotted lines marking the 1 ppm and 1 ppb reference levels (left); wall-clock speed-up of `tpeanuts` over legacy PEANUTS for the Earth-propagation probability, as a function of energy-grid and nadir-grid size (right).

5.1.3 nuSQuIDS validation and performance comparison

Figure 3 (left) summarises the maximum absolute and relative differences between `tpeanuts` and `nuSQuIDS` across four validated configurations: vacuum, atmosphere, solar and the composed solar-to-Earth-detection pipeline. The vacuum and atmosphere modules agree to within 10^{-8} – 10^{-7} in relative terms, comfortably below the 1 ppb reference line, as expected for two independent, essentially exact solvers of the same closed propagation problem. The solar module and the composed solar-Earth-detection pipeline instead show markedly larger discrepancies,

at the percent ($\sim 10^{-2}$) and $\sim 10\%$ level respectively, several orders of magnitude above the vacuum/atmosphere agreement.

This pattern is consistent with the validation strategy of section 3.1.2: unlike the vacuum and atmosphere cases, the solar and solar-Earth-detection observables depend on production-averaged, incoherent mass-eigenstate weights and on the treatment of coherence loss between the Sun and the Earth, both of which **tpeanuts** inherits directly from the PEANUTS convention (section 2.1) but which **nuSQuIDS** may implement through different internal approximations. The comparison against **nuSQuIDS** therefore isolates exactly the kind of convention-dependent discrepancy that a comparison against PEANUTS alone, which shares the same conventions, cannot expose.

Figure 3 (right) shows the wall-clock speed-up of **tpeanuts** over **nuSQuIDS** for the integrated solar-detector probability calculation, as a function of the energy-grid and nadir/ $\cos\theta_z$ -grid sizes used in the batched evaluation. The speed-up grows monotonically with grid size, from $\sim 63\times$ for the smallest grid considered (2×2) to over $1100\times$ for the largest (32×32), directly confirming the scaling behaviour anticipated in section 3.1.3: because **nuSQuIDS** evaluates one solver object per energy/angle point while **tpeanuts** evaluates the full grid in a single batched call, the relative advantage of the tensor-based formulation increases with the number of configurations evaluated per call.

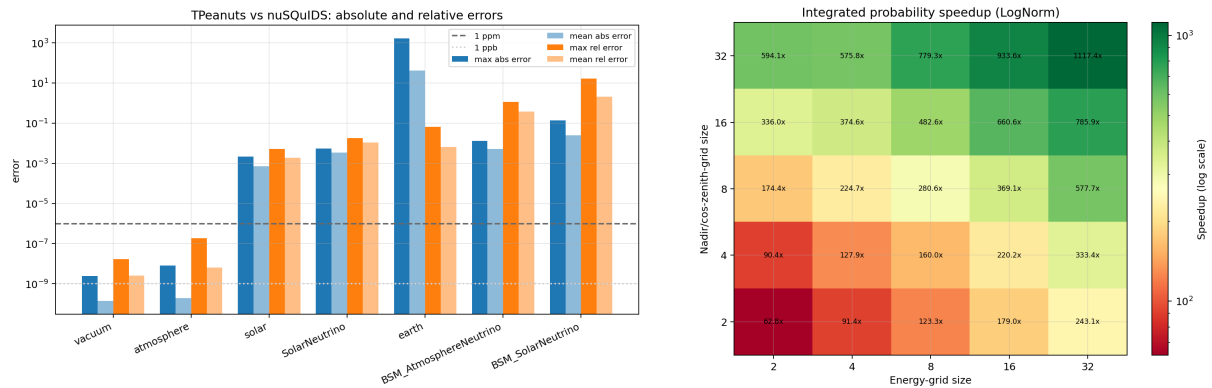


Figure 3: **nuSQuIDS** validation and performance: maximum and mean absolute and relative differences between **tpeanuts** and **nuSQuIDS**, by validated configuration, dashed and dotted lines marking the 1 ppm and 1 ppb reference levels (left); wall-clock speed-up of **tpeanuts** over **nuSQuIDS** for the integrated solar-detector probability, as a function of energy-grid and nadir-grid size (right).

5.2 Simulation analysis

Having established in section 5.1 that the translated implementation is numerically reliable and computationally useful, this subsection exercises the validated engine on complete vacuum, solar- and atmospheric-neutrino propagation pipelines, from production through Earth regeneration, rather than a single isolated probability, checking each result against the qualitative or quantitative behaviour expected from the standard oscillation phenomenology. Section 5.3 then goes further, reproducing a series of specific, well-known experimental signatures with the same engine.

5.2.1 Vacuum neutrino

Figure 4 maps the vacuum survival probability $P(\nu_e \rightarrow \nu_e)$ and the vacuum appearance probability $P(\nu_\mu \rightarrow \nu_e)$ over the full (L, E) plane, L from 1 to 1.5×10^4 km and E from 0.1 to 10 GeV, the baseline/energy range spanning reactor, accelerator and atmospheric configurations within a

single, common map. Both channels are governed by the same oscillatory phase $1.267 \Delta m_{31}^2 L/E$, so the diagonal fringes of P_{ee} and $P_{\mu e}$ run parallel and are exactly complementary at leading order: P_{ee} is close to unity (light) wherever $P_{\mu e}$ is close to zero (light), and P_{ee} dips towards its first oscillation minimum (dark blue) exactly where $P_{\mu e}$ peaks near its maximum appearance amplitude (dark red on the right), consistent with two-flavour-dominated conversion at fixed L/E once the sub-leading solar splitting is neglected.

The fine chequerboard structure visible at the largest L/E , most pronounced in the bottom-right corner of both panels, is the beat pattern between the atmospheric and solar mass splittings rather than numerical noise: at those baselines and energies the solar phase $\Delta m_{21}^2 L/E$ has itself completed many cycles, so the combined pattern aliases against the finite grid resolution used to render the map. Every pixel is computed from the full three-flavour vacuum evolution operator, so this map is a single, direct consistency check that the implementation reproduces the correct L/E scaling and relative amplitude of survival and appearance before any matter effect is introduced, complementing the CP-violation check of figure 12 below, which isolates the phase structure of the same vacuum evolution operator rather than its L/E scaling.

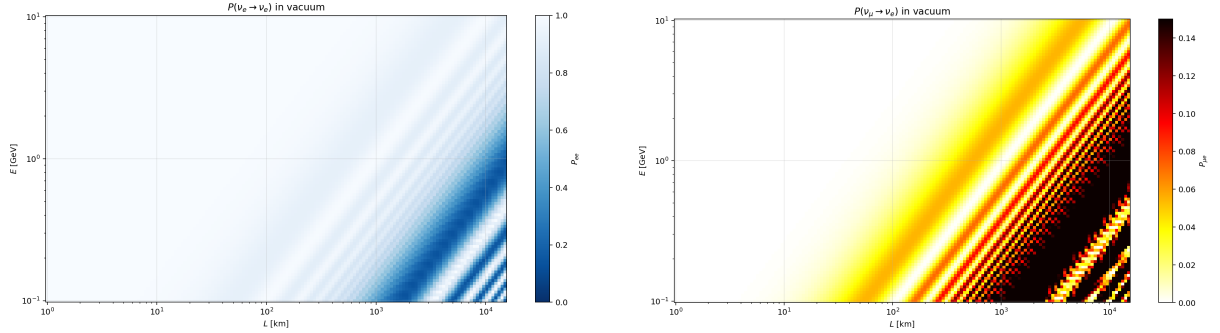


Figure 4: Vacuum oscillation probabilities over the (L, E) plane: survival $P(\nu_e \rightarrow \nu_e)$ (left) and appearance $P(\nu_\mu \rightarrow \nu_e)$ (right).

5.2.2 Solar neutrino

Figure 5 shows the Earth-regeneration probability $P(\nu_\odot \rightarrow \nu_\alpha; E)$ for incoherent solar ${}^8\text{B}$ mass-basis weights propagated along four representative nadir-angle trajectories, crossing the inner core, outer core, mantle and crust only, against the no-Earth vacuum reference. Below ~ 4 MeV every trajectory overlaps the vacuum curve almost exactly, since the day–night regeneration mechanism scales with the matter path length and electron density integrated along the trajectory, both small compared with the vacuum oscillation length at these energies. Above ~ 6 MeV the curves separate visibly from vacuum and from each other, with the inner-core-crossing trajectory showing the largest departure, consistent with its longer, denser matter path driving the strongest regeneration, the same mechanism quantified through the day–night asymmetry in section 5.3.1 below. The three flavour channels remain unitary at every energy and trajectory, $\sum_\alpha P(\nu_\odot \rightarrow \nu_\alpha) = 1$, confirming that Earth regeneration redistributes flavour content without any leakage.

Figure 6 exercises the solar propagation pipeline with all solar production channels combined (pp , pep , hep , ${}^7\text{Be}$, ${}^8\text{B}$, ${}^{13}\text{N}$, ${}^{15}\text{O}$, ${}^{17}\text{F}$), giving the total flavour flux $\Phi_\alpha^{\text{tot}}(E, \eta_{\text{fix}})$ reaching the Earth’s surface at a fixed nadir angle. The ν_e flux falls steadily from $\sim 3.5 \times 10^{10}$ to $\sim 2 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ as the survival probability decreases with energy, while ν_μ and ν_τ grow

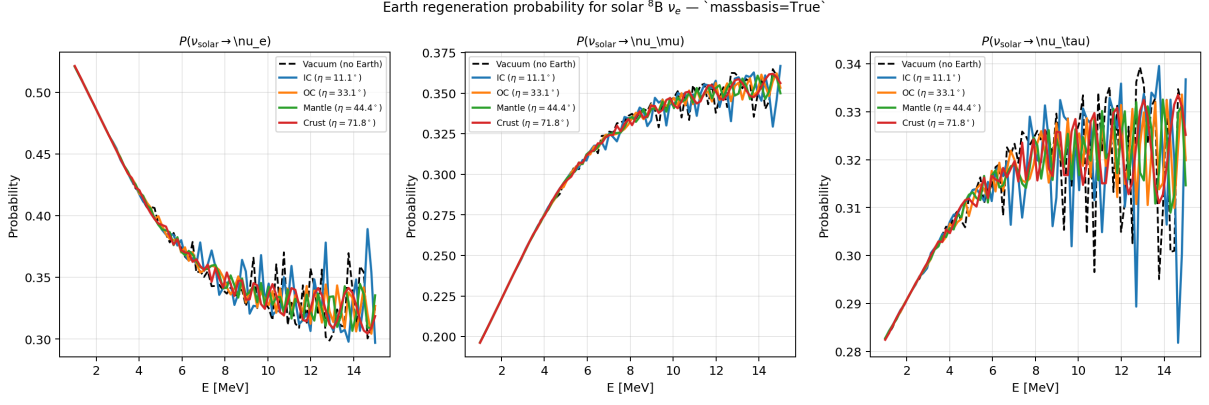


Figure 5: Earth-regeneration probability for solar ^8B neutrinos produced as incoherent mass-basis weights, $P(\nu_\odot \rightarrow \nu_\alpha; E)$, for four representative nadir-angle trajectories crossing the inner core, outer core, mantle and crust, compared with the no-Earth vacuum reference.

correspondingly and converge above ~ 9 MeV, consistent with the near-equal μ/τ appearance implied by the PMNS structure in the solar sector.

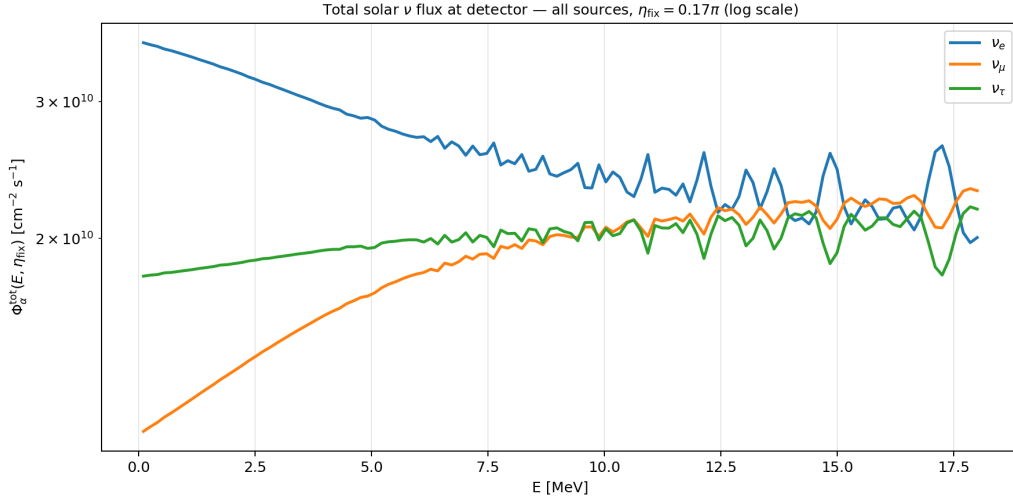


Figure 6: Total solar-neutrino flux reaching the Earth's surface, $\Phi_\alpha^{\text{tot}}(E, \eta_{\text{fix}})$, summed over all solar production channels, at fixed nadir angle $\eta_{\text{fix}} = 0.17\pi$.

Isolating the ^8B channel and exercising the nadir-angle exposure integration for four representative surface latitudes gives three flavour spectra that are, to the resolution of the plot, indistinguishable across all four: annual averaging over the day/night nadir-angle exposure smooths out the latitude-dependent matter effects seen in figure 6, as expected from the small latitude differences considered.

5.2.3 Atmospheric neutrino

The atmospheric detection pipeline (section 4.3) requires, as a boundary condition, the production-height distribution of the parent flux at each energy and angle. Figure 7 compares this distribution, $f(h|E, \alpha)$, for ν_μ at a fixed zenith angle ($\alpha = 31.8^\circ$) as generated by MCEq (Fedynitch et al., 2015) and by the tabulated Honda/HKKM parametrisation (Honda et al., 2007). Both place the bulk of the production between ~ 10 and 30 km, peaking around 15 km and shifting slightly higher at higher energy; the visibly discretised Honda distribution reflects its coarser production-height binning, but the two agree closely in shape and normalisation.

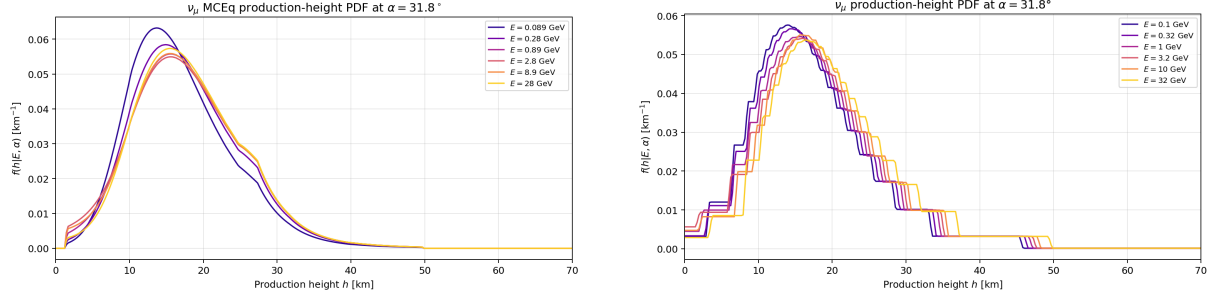


Figure 7: ν_μ production-height probability density, $f(h|E, \alpha)$, at fixed zenith angle $\alpha = 31.8^\circ$, generated with MCEq (left) (Fedynitch et al., 2015) and with the Honda/HKKM parametrisation (right) (Honda et al., 2007).

Figure 8 shows the corresponding surface oscillation probabilities $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_\alpha; E, h)$ for an antimuon-neutrino primary at fixed zenith angle $\theta = 18.2^\circ$, as a function of energy and production height, before folding in the production-height distribution of figure 7 or propagating further through the Earth. Survival dominates almost everywhere above ~ 0.3 GeV, $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_\mu) \gtrsim 0.8$, while the appearance channels $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_e)$ and $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_\tau)$ are confined to a narrow band below ~ 0.3 GeV that widens with production height, tracing out the longer atmospheric path length available to neutrinos produced higher up, exactly the L/E -driven onset of oscillation expected before any Earth matter effect is introduced.

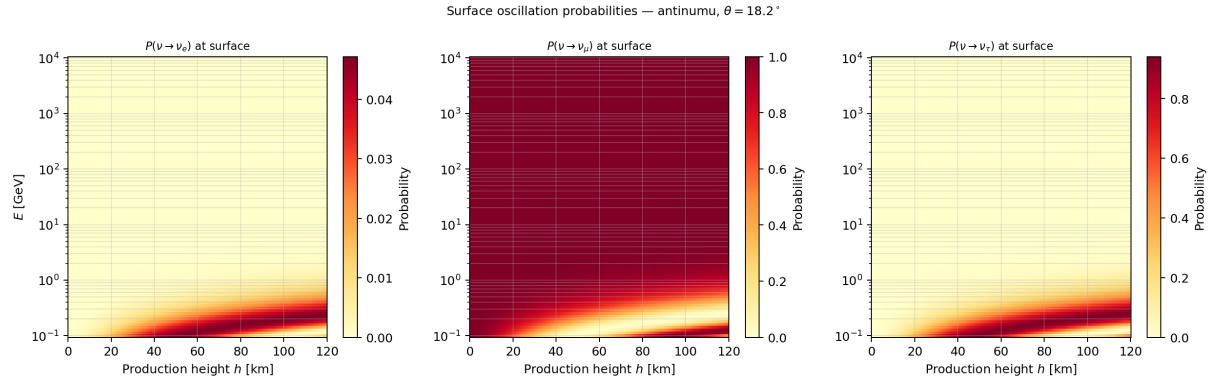


Figure 8: Surface oscillation probabilities $P(\bar{\nu}_\mu \rightarrow \bar{\nu}_\alpha; E, h)$ for an antimuon-neutrino primary at fixed zenith angle $\theta = 18.2^\circ$, as a function of energy and production height.

Figure 9 shows the resulting detector-level ν_μ survival probability, $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$, after propagation through the atmosphere and Earth and averaging over the production-height distribution of figure 7. The pipeline reproduces the expected atmospheric ν_μ disappearance: survival close to unity at high energy ($E \gtrsim 100$ GeV), decreasing towards ~ 0.1 – 0.2 at sub-GeV energies and large zenith angles, confirming that the height-integration and flavour-summation stages combine correctly with the oscillation engine.

5.3 Reproduction of experimental results

Building on the validated propagation pipelines of section 5.2, this subsection reproduces a series of specific, well-known experimental and phenomenological signatures, the solar day–night asymmetry, the adiabatic approximation and the MSW resonance built on it, CP violation, atmospheric oscillograms, and Earth matter tomography, each checked against a published measurement or an analytic expectation rather than only a qualitative pattern.

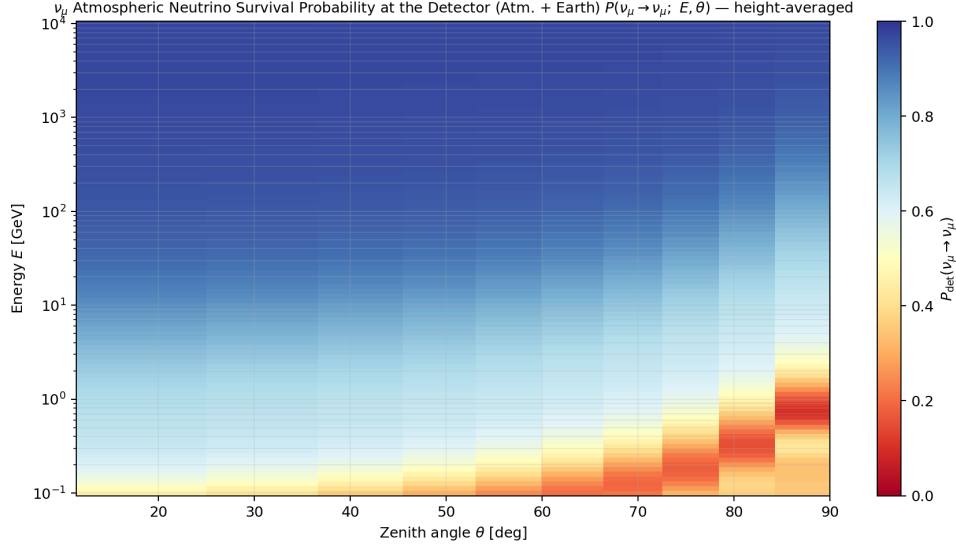


Figure 9: Height-averaged, detector-level ν_μ survival probability $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$, combining atmospheric and Earth propagation.

5.3.1 Solar day–night asymmetry

The regeneration of solar ν_e inside the Earth during night-time observation is quantified by the day–night asymmetry $A_{\text{DN}} = 2(P_{ee}^{\text{day}} - P_{ee}^{\text{night}})/(P_{ee}^{\text{day}} + P_{ee}^{\text{night}})$, computed here for the ^8B flux using the NuFIT 5.2 normal-ordering parameters (Esteban et al., 2020). The flux-weighted, energy-integrated asymmetry $A_{\text{DN}}^{\text{int}}$, obtained for the Super-Kamiokande site, is compared directly against the combined Super-Kamiokande measurement: the predicted -2.69% falls within the experimental 1σ band, $(-3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$ (Renshaw et al., 2014), confirming that the solar detection pipeline reproduces the observed regeneration signal within present experimental precision.

A more detailed, detector-specific comparison additionally accounts for two effects relevant to the Super-Kamiokande measurement: its 3.5 MeV electron-recoil cut corresponds to a higher effective neutrino-energy threshold, $E_\nu^{\text{eff}} \approx 5.5$ MeV, and the observed rate is weighted by the full elastic-scattering cross section rather than by the neutrino spectrum alone. Table 4 shows that both corrections push the predicted asymmetry closer to the measured value, from -2.69% to -3.20% , in better agreement with the Super-Kamiokande measurement and consistent with the mild ($\sim 1.3\sigma$) tension already present between that measurement and the independent LMA-MSW theoretical range.

Table 4: Day–night asymmetry $A_{\text{DN}}^{\text{int}}$ at Super-Kamiokande: as computed ($E_\nu > 3.5$ MeV threshold, spectral weight only) and with the two Super-Kamiokande-specific corrections applied, compared against the independent LMA-MSW prediction and the combined Super-Kamiokande measurement.

Detector	tpeanuts	tpeanuts-sk	LMA std.	Published
Super-K	-2.69%	-3.20%	-2.0% to -1.7%	$(-3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$

5.3.2 Adiabatic approximation

The validity of the adiabatic approximation is assessed through the adiabaticity parameter $\gamma(\rho)$, whose value must remain far above the non-adiabatic onset, $\gamma = 1$, for the adiabatic MSW formalism to apply (Haxton, 1986; Parke, 1986). At representative energies from 5 to 15 MeV

(the least favourable case), γ at the resonance layer ranges from $\sim 6.4 \times 10^3$ down to $\sim 1.7 \times 10^3$, several orders of magnitude above threshold even at its lowest, confirming that the adiabatic approximation is an excellent description across the full solar-neutrino energy range and that the Landau–Zener corrections of section 3.1.2 remain small. Figure 10 shows $\gamma(\rho)$ across the full solar production region for several representative energies: it stays orders of magnitude above the $\gamma = 1$ onset everywhere except in a narrow region near the solar surface, decreasing monotonically with energy as expected from the weaker effective coupling at higher E .

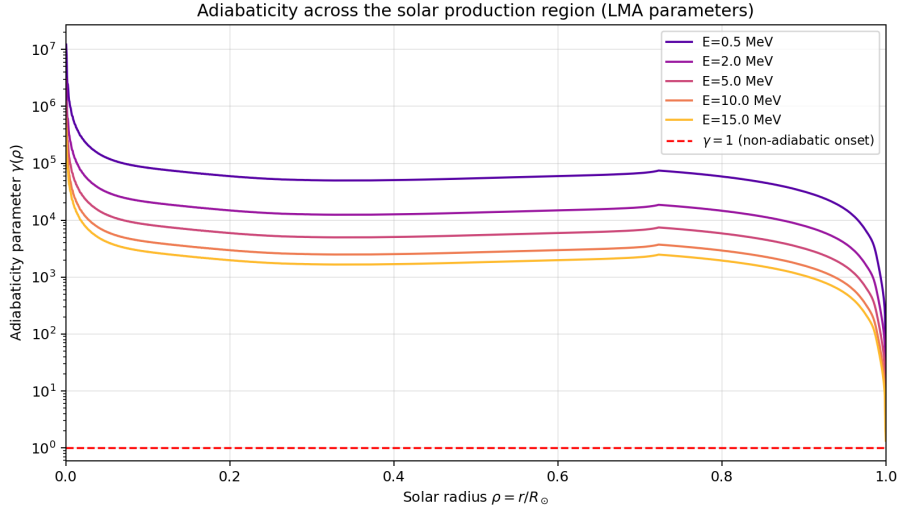


Figure 10: Adiabaticity parameter $\gamma(\rho)$ across the solar production region, for five representative energies, against the non-adiabatic onset $\gamma = 1$.

5.3.3 MSW resonance

The MSW resonance condition, $V_k(\rho) = \cos 2\theta_{12}$, is located along the solar density profile for several neutrino energies. At low energy ($E \lesssim 1.5$ MeV) the potential never reaches the resonance condition within the Sun, so pp -like neutrinos propagate quasi-adiabatically without crossing resonance; as the energy increases, the resonance radius ρ_{res} moves outward, from $\rho_{\text{res}} = 0.032$ at 2 MeV to $\rho_{\text{res}} = 0.282$ at 15 MeV, reproducing the expected energy dependence of the MSW resonance layer. Figure 11 shows the local survival probability $P_{ee}(\rho, E)$ across the production region. At low energy, P_{ee} remains close to the vacuum-averaged value, $\sum_i |U_{ei}|^4 = 0.553$, throughout the Sun; at high energy, P_{ee} starts near the adiabatic matter-eigenstate limit, $|U_{e2}|^2 = 0.296$, at the production point and relaxes smoothly to the vacuum-averaged value as the neutrino exits the Sun, exactly as expected from adiabatic MSW conversion.

5.3.4 CP Violation and Mass Ordering

Figure 12 shows the appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a long-baseline configuration ($L = 1300$ km) for the NuFIT 5.2 best-fit CP phase, $\delta_{\text{CP}} = 197^\circ$, and for maximal CP violation, $\delta_{\text{CP}} = -90^\circ$, together with the CP asymmetry $A_{\text{CP}} = P_{\mu e} - \bar{P}_{\mu e}$. The asymmetry is largest at low energy, where the genuinely CP-odd term dominates, and vanishes at the first oscillation node near $E \sim 1$ GeV, consistent with the expected L/E dependence.

The reactor-antineutrino survival probability illustrates why establishing the neutrino mass ordering is comparatively difficult. Figure 13 compares $P(\bar{\nu}_e \rightarrow \bar{\nu}_e)$ at a Daya-Bay-like baseline ($L = 50$ km) for the normal and inverted orderings: the two curves are visually almost indis-

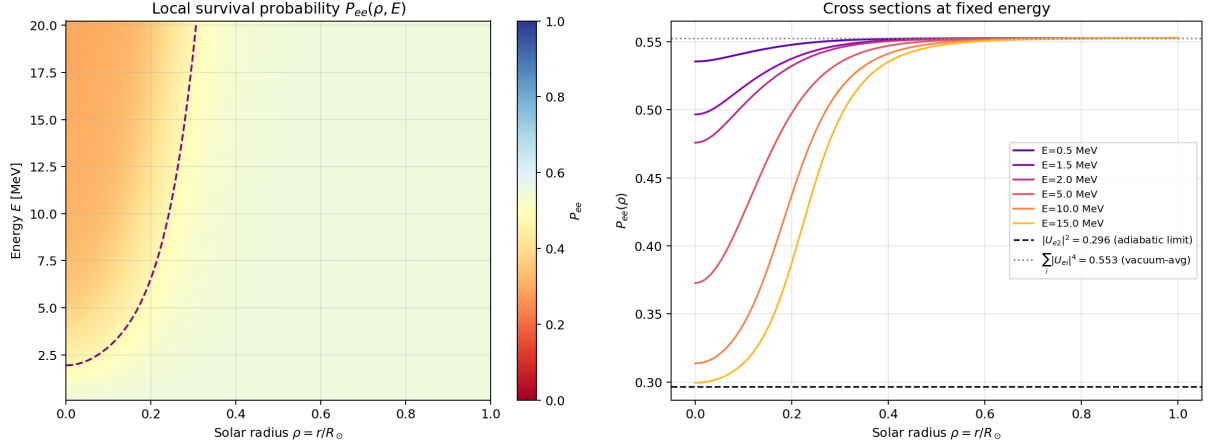


Figure 11: Local survival probability $P_{ee}(\rho, E)$ across the solar production region (left), and radial cross-sections at fixed energy (right), showing the transition between the adiabatic matter-eigenstate limit and the vacuum-averaged value.

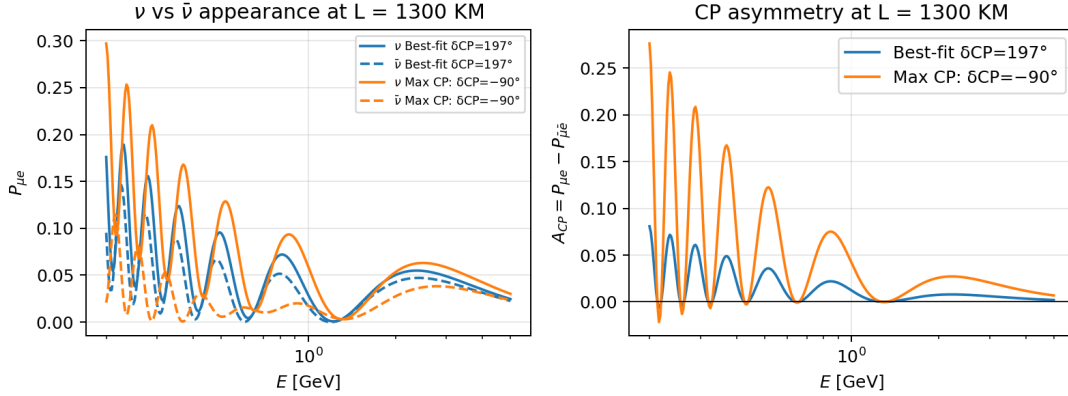


Figure 12: Appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a long-baseline configuration, $L = 1300$ km (left) and the resulting CP asymmetry A_{CP} (right), for the best-fit and maximal-CP-violation values of δCP .

tinguishable, since Δm_{31}^2 and Δm_{32}^2 differ only by the small solar splitting Δm_{21}^2 , and their difference (right panel) oscillates rapidly between roughly $\pm 3\%$ rather than producing a single, easily resolved offset. Extracting the mass ordering from this observable therefore requires resolving the relative phase between the two split oscillation frequencies across a broad energy range at sub-percent energy resolution, rather than detecting one large-amplitude effect, precisely the strategy pursued by medium-baseline reactor experiments such as JUNO.

5.3.5 Atmospheric neutrino oscillograms

Figure 14 shows the atmospheric ν_μ survival-probability oscillogram in the $(\cos \theta_z, E)$ plane, computed through the PREM Earth density profile (Dziewonski & Anderson, 1981). The characteristic vacuum oscillation dip is verified against the analytic first-minimum condition, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$ (Ashie et al., 2004), and against a set of benchmark observables summarised in table 5, spanning the atmospheric mass splitting and mixing angle, the vacuum L/E dip position, the multi-GeV up/down asymmetry, the energy of maximal mantle-crossing suppression, and the maximum appearance probability associated with the mantle MSW resonance.

5.3.6 Earth matter tomography

Figure 15 shows the derivative of the electron-neutrino appearance probability with respect to the nadir angle, $|dP_e/d\eta|$, which develops sharp, localised features precisely at the PREM shell

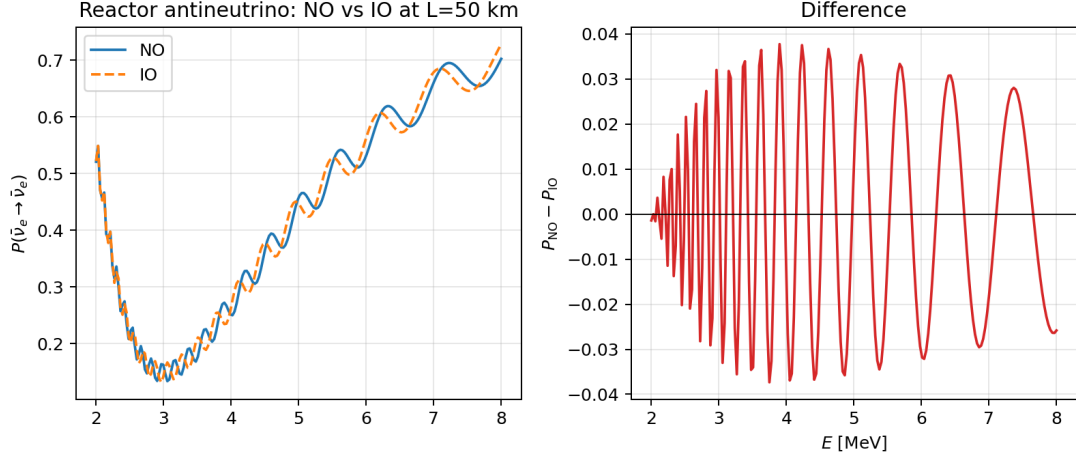


Figure 13: Reactor-antineutrino survival probability $P(\bar{\nu}_e \rightarrow \bar{\nu}_e)$ at $L = 50$ km for the normal and inverted mass orderings (left), and their difference (right), showing the fine, rapidly oscillating imprint the mass ordering leaves on this observable.

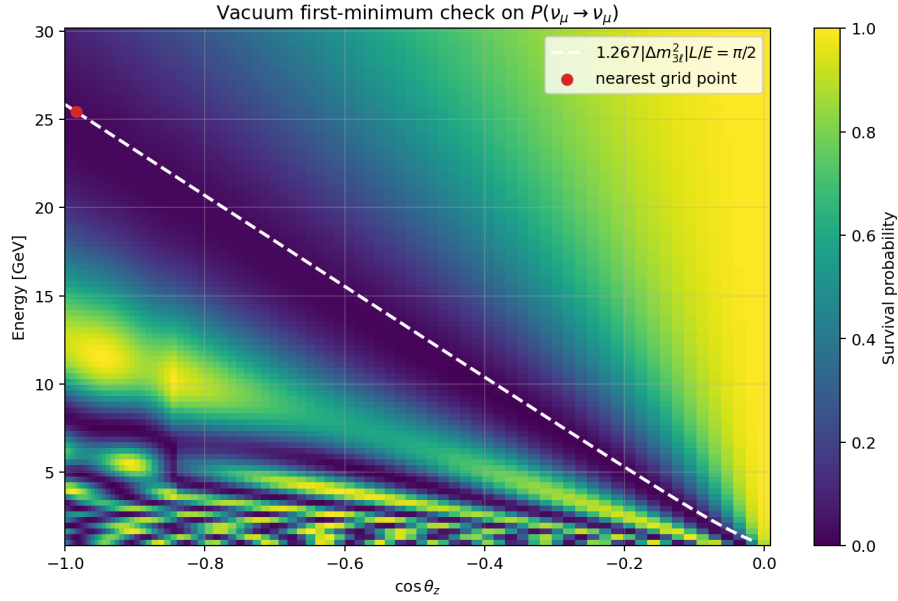


Figure 14: Atmospheric $P(\nu_\mu \rightarrow \nu_\mu)$ oscillogram in the $(\cos \theta_z, E)$ plane. The dashed line marks the analytic first vacuum-oscillation minimum, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$; the red marker shows the nearest grid point.

Table 5: Benchmark observables reproduced from the atmospheric oscillogram, compared with published experimental results.

Observable	Experiment	Reference
$ \Delta m_{3\ell}^2 $ [eV ²]	$[1.9, 3.0] \times 10^{-3}$	Wester et al. (2024), 90% CL (Super-Kamiokande Collaboration, Wester, et al., 2024)
$\sin^2(2\theta_{13})$	> 0.90	Wester et al. (2024), 90% CL (Super-Kamiokande Collaboration, Wester, et al., 2024)
L/E [km/GeV]	≈ 500	Ashie et al. (2004) (Ashie et al., 2004)
A_μ	≈ -0.32	Fukuda et al. (1998) (Fukuda et al., 1998)
Max-3.28 suppression [GeV]	3–30 GeV range	Aartsen et al. (2013), DeepCore (IceCube Collaboration, Aartsen, et al., 2013)
max $B(5904 \rightarrow \nu_e)$	mantle MSW resonance	Aiello et al. (2024), KM3NeT/ORCA (KM3NeT Collaboration, Aiello, et al., 2024)

boundaries (Dziewonski & Anderson, 1981), a pronounced spike at the core–mantle density jump and a regular oscillatory pattern for purely mantle-crossing trajectories, confirming that the propagation correctly resolves discontinuous density profiles, the basic requirement for neutrino tomography.

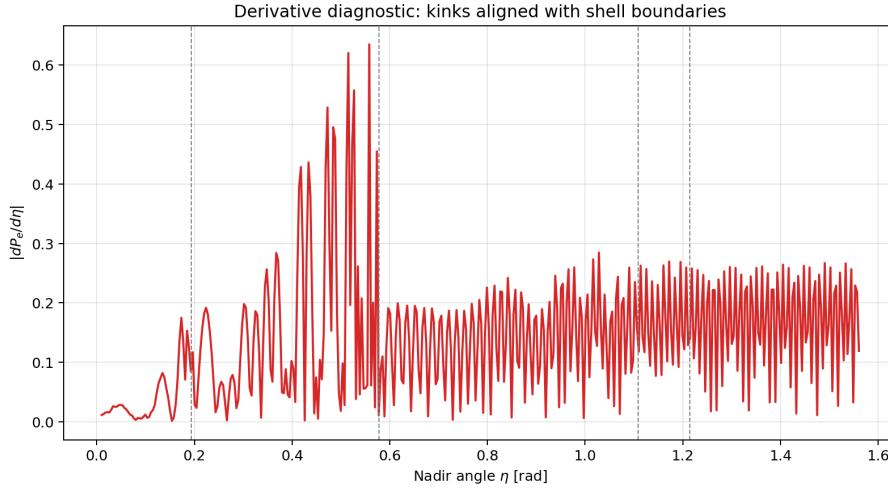


Figure 15: Derivative of the electron-neutrino appearance probability with respect to the nadir angle, $|dP_e/d\eta|$; the dashed vertical lines mark the PREM shell boundaries.

5.4 BSM extensions

5.4.1 NSI

The propagation-NSI formalism introduced in section 2.3.1 is validated by evaluating the solar ^8B survival probability under several representative NSI presets drawn from the literature and from the projected sensitivity of upcoming experiments. Figure 16 compares the Standard Model prediction (no NSI) against the diagonal bound of Biggio et al. (2009), an off-diagonal $\varepsilon_{e\tau} = 0.15$ coupling representative of DUNE’s projected sensitivity, a combined $\varepsilon_{ee} + \varepsilon_{e\tau}$ scenario representative of Hyper-Kamiokande, a purely off-diagonal $\mu\tau$ coupling bounded by IceCube, and the global-fit values of Esteban et al. (2018). All presets converge to the same vacuum-averaged and adiabatic limits as the Standard Model at low and high energy, respectively, but distort the survival probability appreciably in the intermediate MSW transition region, by up to ~ 2 percentage points at $E = 10$ MeV; the largest distortions come from the global-fit and combined Hyper-K-like presets, while the purely $\mu\tau$ off-diagonal coupling leaves P_{ee} exactly unchanged, since it does not couple directly to the electron-flavour survival probability at this order.

An independent scan of $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ over a representative range (-0.3 to $+0.3$), with all other NSI parameters at zero, produces a clear, monotonic and oppositely signed distortion of the high-energy survival probability ($E \gtrsim 3$ MeV) relative to the Standard Model for each coupling (figure 17), confirming that the implementation correctly resolves both the sign and the magnitude dependence of each independent off-diagonal coupling.

5.4.2 Sterile neutrino

The $3 + 1$ sterile extension of section 2.3.2 is validated against analytic expectations for three presets (a null-mixing reference, a Giunti (2017) global-fit-motivated best fit, and an IceCube (2020) benchmark), all propagated in vacuum. Table 6 summarises the confirmed, directly

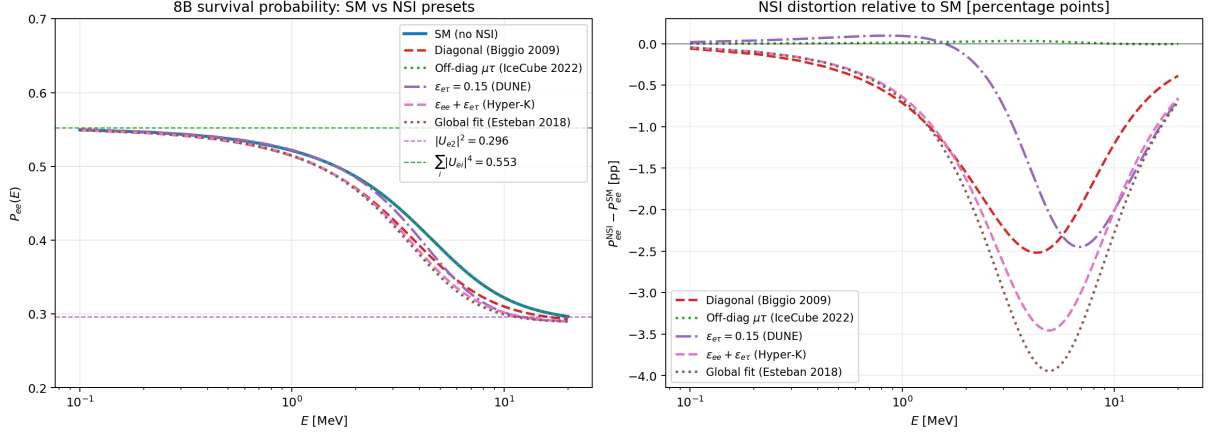


Figure 16: ^8B survival probability $P_{ee}(E)$ under the Standard Model and several representative NSI presets (left), and the resulting distortion relative to the Standard Model, in percentage points (right).

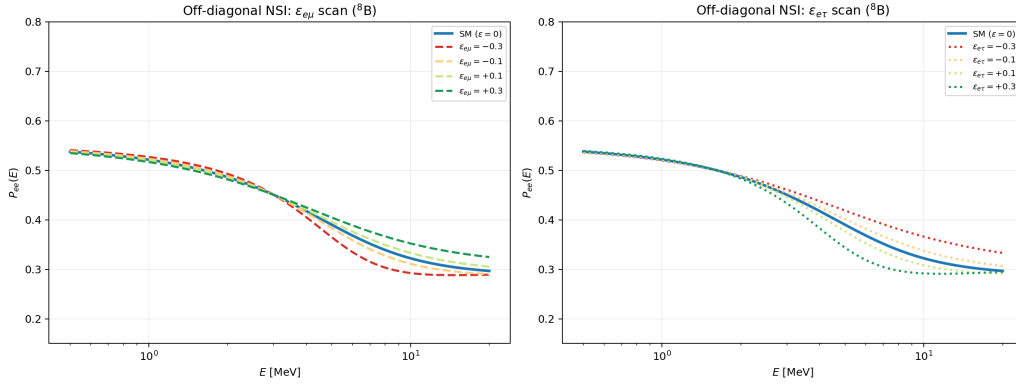


Figure 17: Distortion of the ^8B survival probability $P_{ee}(E)$ relative to the Standard Model, from independent scans of $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ over -0.3 to $+0.3$, with all other NSI parameters fixed to zero.

recorded results: the null-mixing preset recovers the Standard-Model limit to floating-point precision in the PMNS submatrix, the Hamiltonian and the vacuum probability; unitarity, $\|U_4 U_4^\dagger - I_4\|$, holds to $\sim 10^{-16}$ for all three presets; and, under the CC-only default (section 2.3.2, no neutron-density profile supplied), the sterile state is confirmed to receive no matter potential, both its row and column of the matter Hamiltonian vanishing identically. A dedicated CP-phase scan at LSND-like kinematics finds a maximal appearance asymmetry of only $|A_{CP}| \sim 10^{-5}$, consistent with the standard result that appreciable $3 + 1$ CP violation requires simultaneous sensitivity to two independent mass-squared splittings, not available at a single short baseline.

Table 6: Confirmed $3 + 1$ sterile-extension validation results.

Check	Result
SM-limit PMNS/Hamiltonian agreement (null mixing)	exact (0 to printed precision)
Unitarity $\ U_4 U_4^\dagger - I_4\ _F$ (three presets)	$1.5\text{--}3.0 \times 10^{-16}$
Total vacuum probability $\sum_\beta P_\beta$	1.00000000, max. dev. 3.3×10^{-16}
Sterile row/column of matter Hamiltonian	exactly 0
Max. appearance CP asymmetry (LSND kinematics)	0.001% at $\delta_{14} = 108.9^\circ$

The kinematic and phenomenological checks below are taken from companion validation notebooks. For ν_e , the two-flavour disappearance approximation is exact to 10^{-14} in the idealised limit, its realistic-mixing deviation fully explained by the naive-versus-exact $|U_{e4}|^2$ convention; for ν_μ , a genuine residual of $\mathcal{O}(10^{-2})$ survives even after this correction. Mapping the three presets

onto reactor, Gallium, LSND/MiniBooNE and IceCube kinematics reproduces the well-known tension of the minimal $3 + 1$ model: combining the reactor and IceCube disappearance bounds constrains the effective appearance amplitude to $A_{e\mu}^{\text{eff}} \lesssim 0.003$, below the $0.003\text{--}0.03$ required by LSND, a global tension $> 3\sigma$ consistent with the literature (Böser et al., 2020). No inference notebook in the current implementation fits real short-baseline data with the sterile extension; this is left as future work.

5.4.3 LMA-Dark degeneracy

Figure 18 reproduces the LMA-Dark degeneracy discussed in section 3.3, comparing the day–night asymmetry predicted at fixed nadir angle ($\eta = 30^\circ$, 1 km depth) under three scenarios: the standard LMA solution ($\varepsilon_{ee} = 0$); a sign-flip demonstration in which only the Earth-matter NSI coupling is switched on ($\varepsilon_{ee} = -2$); and the full LMA-Dark approximation, obtained by simultaneously transforming $\theta_{12} \rightarrow \pi/2 - \theta_{12}$ and $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$. The standard-LMA solution gives $A_{\text{DN}} \approx +4.8\%$ (here A_{DN} is defined with the opposite sign convention to section 5.3.1, positive when the night rate exceeds the day rate), broadly consistent with the combined Super-Kamiokande measurement $A_{\text{DN}} = (3.2 \pm 1.1_{\text{stat}} \pm 0.5_{\text{syst}})\%$ (Renshaw et al., 2014); switching on only the Earth-matter NSI coupling flips the sign, $A_{\text{DN}} \approx -1.6\%$; and the full LMA-Dark approximation gives $A_{\text{DN}} \approx -2.9\%$, the wrong sign relative to the Super-Kamiokande band and disfavoured by it at the $\sim 3\sigma$ level. Unlike the vacuum mixing angle alone, which is degenerate under the LMA-Dark transformation, the Earth-matter day–night asymmetry breaks the degeneracy at this coupling strength, illustrating why complementary observables beyond the vacuum survival probability are needed to resolve it in general.

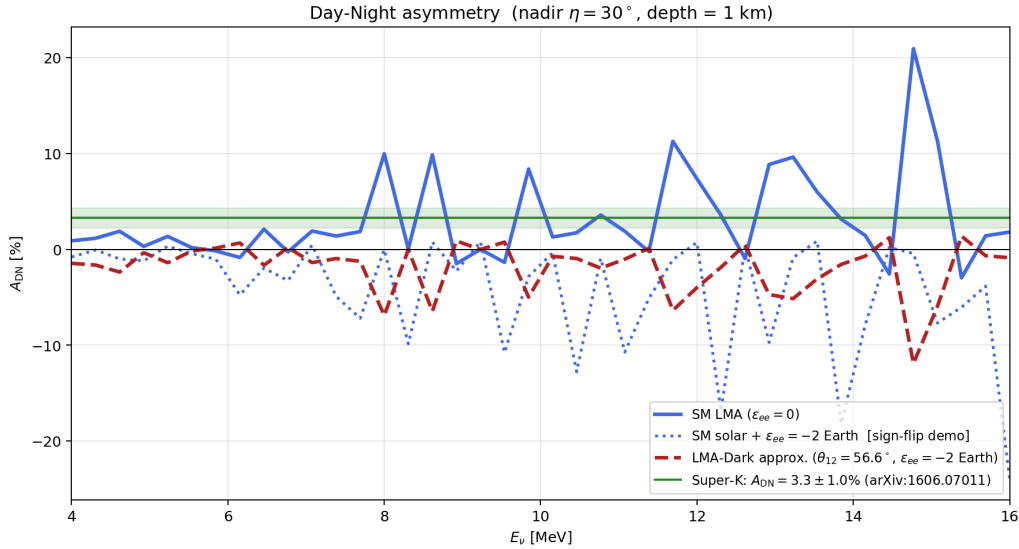


Figure 18: Day–night asymmetry versus energy at fixed nadir angle ($\eta = 30^\circ$, 1 km depth), comparing the standard LMA solution, a sign-flip demonstration with an Earth-only NSI coupling, and the full LMA-Dark approximation, against the combined Super-Kamiokande measurement (Renshaw et al., 2014).

5.5 Inference with real experimental data

5.5.1 Detector-level case study: Daya Bay

The detector and inference layers of section 4.1.1 are validated end to end against the real, published Daya Bay reactor-antineutrino data set: 8 antineutrino detectors (ADs) at three

experimental halls, 6 reactor cores, and the 8-AD-period inverse-beta-decay (IBD) prompt-energy spectra, following the methodology of section 3.4.

The closure test, evaluated at Daya Bay’s own published oscillation parameters with no free normalisation, predicts 3 705 407 IBD counts across all 8 detectors against the 3 114 934 actually observed, a ratio of 1.190: the forward model over-predicts the observed rate by $\approx 19\%$, remarkably uniformly across all 8 detectors. A dedicated normalisation audit traced this gap through every stage of the absolute-rate chain, target mass, efficiency, cross section (independently validated to $\sim 1\%$ against Vogel & Beacom’s own quoted coefficient), and the Huber–Mueller yield, identifying and quantifying its dominant sources, with no unit-conversion error found at any traced stage (a compensating error elsewhere in the chain would not be excluded by this audit alone): part of the gap is the well-known, independently measured Reactor Antineutrino Anomaly ($\sim 5\text{--}6\%$, measured/predicted flux ratio 0.946 ± 0.020 An et al., 2017), and the remainder is consistent with the correlated normalisation systematics Daya Bay’s own combined analysis also carries, precisely why a free ν_{norm} nuisance is included in the fits below rather than fixed to 1.

Comparing the predicted and observed prompt-energy spectra for one near-hall and one far-hall detector shows that both share the same IBD-kinematics-driven shape, with the far-hall spectrum visibly suppressed relative to its no-oscillation expectation, the qualitative signature the measurement is built on. Fitting the full 8-detector, 208-bin spectral shape jointly over $(\theta_{13}, \Delta m_{31}^2, \nu_{\text{norm}})$ was attempted with increasingly rich nuisance freedom (background-rate and energy-scale-non-linearity pulls); none of the three attempts converges to a physical point, and adding energy-scale pulls makes the fit worse, $\hat{\theta}_{13}$ collapsing to 0° , a genuine degeneracy between the oscillation deficit and uncorrelated energy-scale freedom rather than a bug. This is reported as a diagnosed limitation: it demonstrates why real reactor experiments build a full correlated systematic covariance rather than a handful of independent nuisances once sub-percent shape residuals dominate the statistical error.

Two simpler observables, by contrast, converge robustly. Integrating each detector’s spectrum to a single rate and fitting $(\theta_{13}, \nu_{\text{norm}})$ with Δm_{31}^2 fixed gives $\hat{\theta}_{13} = 8.389^\circ \pm 0.139^\circ$ ($\sin^2 2\theta_{13} = 0.08332$), -2.66% relative to Daya Bay’s published 0.08560 (-0.85σ). The near/far ratio fit, mirroring Daya Bay’s own 2012 discovery strategy (An et al., 2012), is the only strategy with genuine joint sensitivity to both parameters, and is the headline result of this case study:

Table 7: Daya Bay near/far ratio fit vs. Daya Bay’s own published best fit and NuFIT 6.1 (Esteban et al., 2024, 2025).

Quantity	<code>tpeanuts</code> (near/far)	Daya Bay published	NuFIT 6.1
$\sin^2 2\theta_{13}$	0.08442	0.0856 (An et al., 2023)	0.0878
θ_{13} [deg]	8.445 ± 0.134	8.506	8.62
Δm_{31}^2 [10^{-3}eV^2]	2.478 ± 0.055	2.528	2.511

A 2-D $-2\ln L$ scan over $(\theta_{13}, \Delta m_{31}^2)$, shown in figure 19, confirms that the fitted point and Daya Bay’s own published point sit well inside the same low- $\Delta(-2\ln L)$ region. As table 7 shows, the near/far fit recovers both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly, within 1–2% of Daya Bay’s own published values and comfortably inside NuFIT’s global-fit range, using `tpeanuts`’s own real multi-reactor flux, order- $1/M$ cross section and IAV+LSNL+Gaussian detector response, not a synthetic data set. This is the clearest demonstration that the detector and inference layers, built

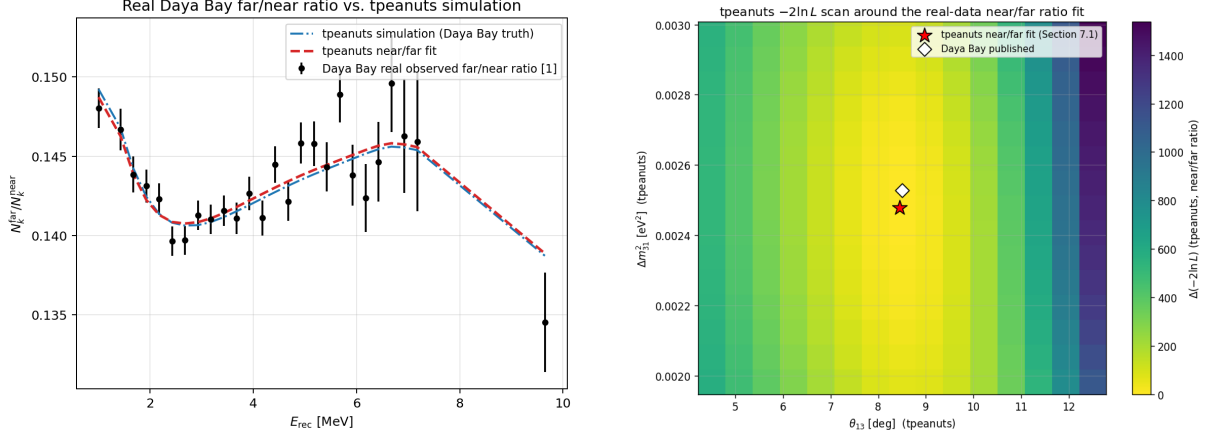


Figure 19: Real Daya Bay far/near per-bin count ratio, compared against the `tpeanuts` simulation at Daya Bay’s own published truth and at the near/far fit (left); `tpeanuts` $-2\ln L$ scan around that fit, in the $(\theta_{13}, \Delta m_{31}^2)$ plane, compared with Daya Bay’s own published best fit (right).

on top of the validated oscillation engine of section 3.1.2, extend `tpeanuts` from fast forward simulation to genuine gradient-based analysis of real experimental data.

5.5.2 Other detector-inference applications

The same detector and inference layers are applied to four further real experiments (table 2), documented in full, with their own validation figures, in the specialised documents referenced in section 5.6:

- **IceCube DeepCore:** a 4-parameter Poisson fit to 9 years of golden atmospheric-event data recovers $\theta_{23} = 45.42^\circ \pm 1.85^\circ$ and $\Delta m_{3l}^2 = (2.395 \pm 0.039) \times 10^{-3} \text{ eV}^2$, a -0.05σ pull on $\sin^2 \theta_{23}$ relative to the collaboration’s published best fit; a dedicated audit of the fitted muon-background normalisation (16% above the published value) rules out an Earth-matter-unitarity origin (a 0.002% effect) and instead traces most of the gap to fixed detector-systematics hypersurfaces (up to +12% for the ν_τ -CC channel) not yet freed as nuisance parameters.
- **SNO phase II:** a fit to the full 391-day, 38-observable salt-phase data set, using the published statistical covariance plus a reconstructed systematic covariance that does not yet include the NC channel’s dedicated systematics, converges to $\theta_{12} = 35.28^\circ \pm 1.61^\circ$, $\Delta m_{21}^2 = (6.10 \pm 34,900) \times 10^{-5} \text{ eV}^2$; θ_{12} sits inside SNO’s own SNO-only 1σ range, while the reported Δm_{21}^2 uncertainty illustrates the Laplace approximation breaking down once the highly correlated reconstructed covariance is included, exactly why the likelihood-scan machinery of section 4.1.1 exists.
- **Borexino:** a direct fit of θ_{12} to the four published $P_{ee}(E)$ points (Borexino Collaboration, Agostini, et al., 2018), starting from a deliberately uninformative guess (maximal mixing), converges to $\theta_{12} = 34.59^\circ \pm 4.15^\circ$, within 0.2σ of NuFIT; the separate, full detector-level elastic-scattering event-rate chain (target, response, cross section) is implemented and unit-tested against the real Nature 2018 low-energy spectrum, but its background model is currently a documented placeholder, so no event-rate fit to real data is yet reported.
- **SNO phase I:** a day/night spectral fit to real published data converges robustly to $\theta_{12} = 34.333^\circ \pm 0.665^\circ$, a 0.86σ pull relative to NuFIT 6.1.

Table 8 summarises these four results against their own published references.

Table 8: `tpeanuts` inference results vs. published values for the four experiments above.

Experiment	Quantity	<code>tpeanuts</code>	Published	Diff. (%)	Reference
IceCube DeepCore	θ_{23} [deg]	45.42 ± 1.85	45.57 ($\sin^2 \theta_{23} = 0.51$)	-0.34	(Abbasi et al., 2023)
IceCube DeepCore	Δm_{3l}^2 [10^{-3} eV ²]	2.395 ± 0.039	2.484	-3.59	(Abbasi et al., 2023)
SNO II	θ_{12} [deg]	35.28 ± 1.61	33.85 ($\tan^2 \theta_{12} = 0.45$)	+4.21	(Aharmim et al., 2005)
SNO II	Δm_{21}^2 [10^{-5} eV ²]	$6.10 \pm 34,900$	$5.0^{+0.2}_{-1.8}$	+22.0	(Aharmim et al., 2005)
Borexino*	θ_{12} [deg]	34.59 ± 4.15	33.76 (NuFIT)	+2.47	(Esteban et al., 2025)
SNO I	θ_{12} [deg]	34.333 ± 0.665	33.76 (NuFIT)	+1.70	(Esteban et al., 2025)

*Direct fit of $P_{ee}(E)$ to Borexino’s own published points, not the full detector-level event-rate fit, which is still pending (see text above).

5.6 Public repository and supplementary material

The full source code of `tpeanuts`, together with the notebooks used for validation, analysis and benchmarking, is publicly available in the following GitHub repository:

<https://github.com/juanramondiaz/tpeanuts>

The repository is organised so that the physical modules are kept separate from the notebooks that exercise them; `notebooks/` is the main entry point for reviewing results and reproducing figures (table 9).

Table 9: Notebook directory tree (paths relative to `notebooks/` in the repository above).

Path	Contents
<code>physics/simulation/{vacuum,solar,earth,atmosphere}</code>	Fundamentals, production/exposure profiles and differential-flux analysis per medium.
<code>physics/experimental_results/{vacuum,solarneutrino,atmosphericneutrino,BSM}</code>	Reproduction of the physical signatures of section 5.3 and table 6.
<code>validation/{nusquids,legacy}</code>	Cross-validation against <code>nuSQuIDS</code> and bit-level validation against legacy PEANUTS.
<code>inference/</code>	Detector-level closure tests and gradient-based fits to real data (Borexino, SNO, Daya Bay, IceCube DeepCore, SNO II; sections 5.5.1 and 5.5.2).
<code>benchmark/</code>	Performance comparison against PEANUTS and <code>nuSQuIDS</code> .
<code>external/{corsika,nusquids,prem,zenodo,reactor/dayabay}</code>	CORSIKA generation and exploratory/data-retrieval notebooks for external inputs.

Full architectural and physics detail beyond the scope of this thesis, module by module, is documented separately in five companion L^AT_EX documents shipped alongside the code (`documentation/`), each with its own figures and, for the inference documents, its own real-data results: `guide` (`core`, `medium`, `pipeline`, `detector.common/interaction/borexino`); `BSM_NSI` and `BSM_sterile` (theory, implementation and notebook results for each BSM extension, table 6); and `inference_dayabay`, `inference_icecube`, `inference_sno_ii` (full detector implementation and real-data results underlying sections 5.5.1 and 5.5.2).

6 Conclusions

This Master’s Thesis has presented the motivation, theoretical basis and development strategy for `tpeanuts`, a modular, publicly available PyTorch framework for neutrino-oscillation phenomenology that grows out of, and substantially extends, the PEANUTS toolkit. The part inherited directly from PEANUTS follows a conservative principle: its physical conventions are preserved while the numerical implementation is modernised for tensor operations, batching and accelerator execution; everything built on top of that foundation is new.

That foundation is the organisation of the PEANUTS numerical core into a PyTorch-oriented design covering PMNS construction, matter potentials, vacuum propagation, matter mixing and solar-neutrino probabilities, together with a validation methodology based on low-level comparisons, intermediate physical quantities and final probabilities. This methodology is essential because speed is useful only if the updated implementation remains physically faithful, and it is the precondition for everything that follows, not the main result in itself.

The central contribution of this work is the modular software architecture described in sections 3.2 and 4.1, which decouples the core oscillation engine from the specific neutrino source, matter-density model and physics scenario being evaluated. This design extends the original PEANUTS code, limited to solar-neutrino propagation, to atmospheric-neutrino detection through the Earth, to multiple interchangeable solar and Earth density models and perturbative treatments of the matter potential, and to beyond-the-Standard-Model extensions (non-standard interactions and sterile neutrinos), all without modifying the underlying propagation engine. The validation and benchmark results of section 5.1 confirm the engine’s numerical reliability; sections 5.2 and 5.3 establish its Standard-Model solar and atmospheric simulations and reproduce a range of well-known physical signatures; and section 5.4 extends that same validated engine to BSM physics, together demonstrating that the modular design is not only a software-engineering convenience but a working foundation for future physics studies.

A second, equally central contribution is extending this engine into a real, data-driven statistical-inference framework: a detector-agnostic forward-folding layer and a gradient-based likelihood/fitting layer, built on PyTorch autograd, connect the same validated oscillation engine to real, published data from five experiments. The Daya Bay case study (section 5.5.1) shows this chain works end to end, a near/far ratio fit recovering both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly within 1–2% of Daya Bay’s published values, and its diagnosed limitations (the full-spectral-shape fit’s non-convergence without a full systematic covariance, the Laplace approximation’s breakdown for SNO II’s Δm_{21}^2) identify concrete next steps rather than open questions.

Taken together, these two contributions, the modular oscillation engine and the detector-inference layer built on top of it, are released publicly (section 5.6) as a documented, installable Python package rather than a closed thesis codebase, intended, given its batched, GPU-capable and differentiable design, to be directly reusable by other researchers working on solar, atmospheric or BSM neutrino phenomenology, not only a record of the results reproduced here.

Future work will focus on these lines: Improving the detector-level analyses for several observatories, mainly through richer correlated systematic treatments; deepening the BSM extensions, in particular connecting the sterile extension to the real-data inference machinery

already validated for the Standard-Model detector layer; extending the current Majorana-neutrino parametrisation, with magnetic-field physics, a transition magnetic moment and the resulting spin-flavour precession Hamiltonian, the channel through which Majorana and Dirac neutrinos actually become experimentally distinguishable; generalising the propagation medium towards an arbitrary-density-profile formulation; and combining the individually validated detector-inference chains into a genuine joint likelihood across experiments.

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